



# **ELL 100 - Introduction to Electrical Engineering**

## **LECTURE 4 : CIRCUIT ANALYSIS**

### **DELTA – STAR TRANSFORMATIONS**

# Outline

- ☐ Introduction
- ☐ Star Connection
- ☐ Delta Connection
- ☐ Delta to Star Transformation
- ☐ Star to Delta Transformation
- ☐ Equivalent Resistance of Circuit
- ☐ Exercise/Numerical Analysis



# INTRODUCTION



Star connections is generally used in long distance transmission lines as insulation requirement is less in star connection.



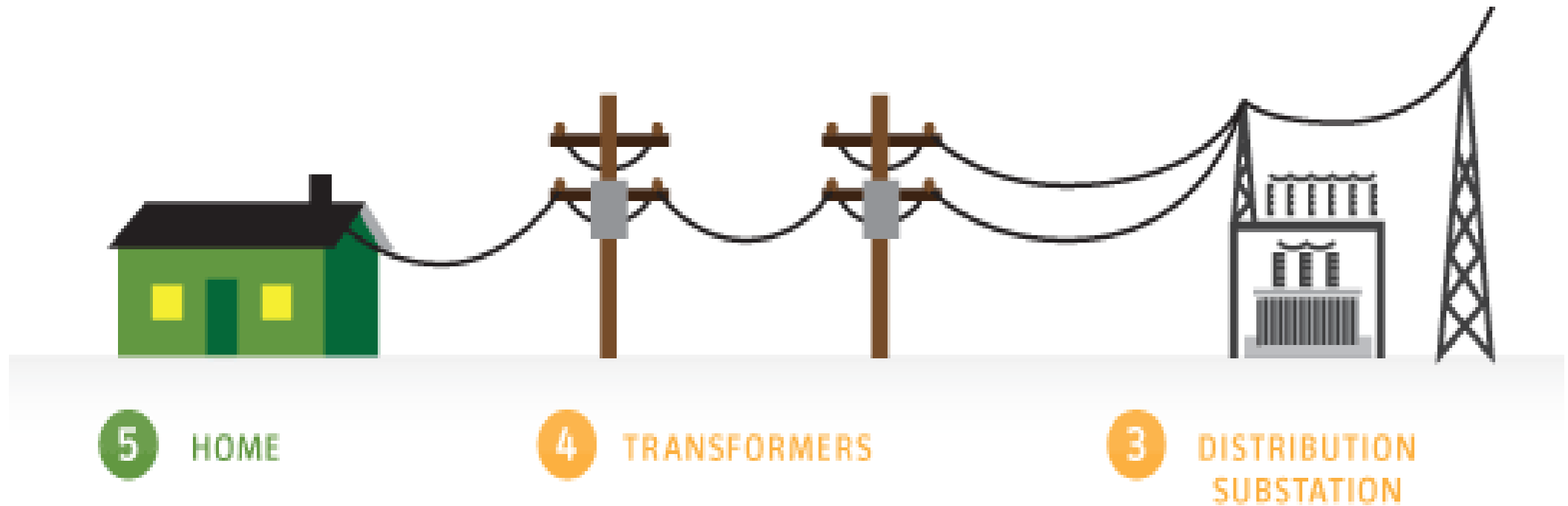
Transmission Network



# INTRODUCTION



Delta connections are generally used in distribution networks for short distances.



Distribution Network





# INTRODUCTION



Alternators and generators are usually star connected.



600 MW Turbo-Generator at power plant

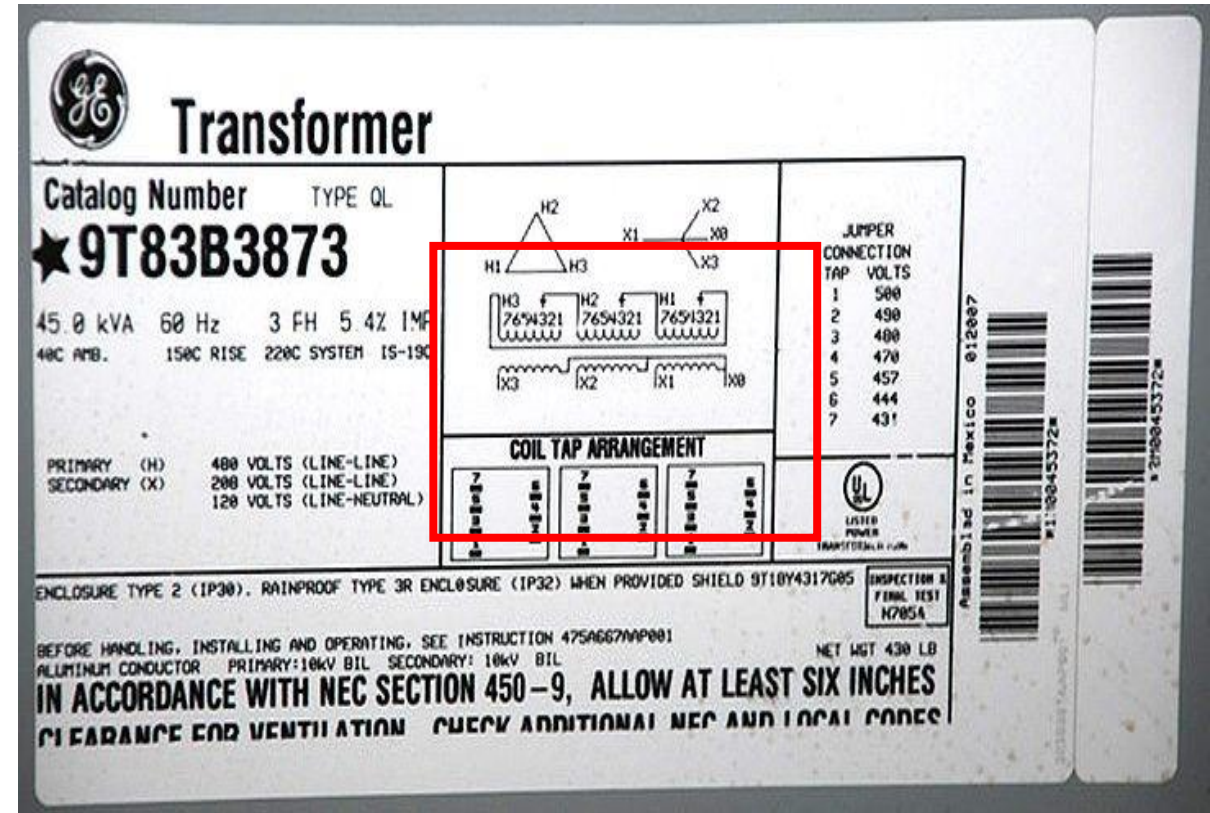




# INTRODUCTION



Transformer windings are connected in Star/Delta Connections.



A Three Phase Transformer with Name Plate



# INTRODUCTION



Generating transformer near to power plant generator are connected in star connection to provide grounding protection.



Generating Transformer

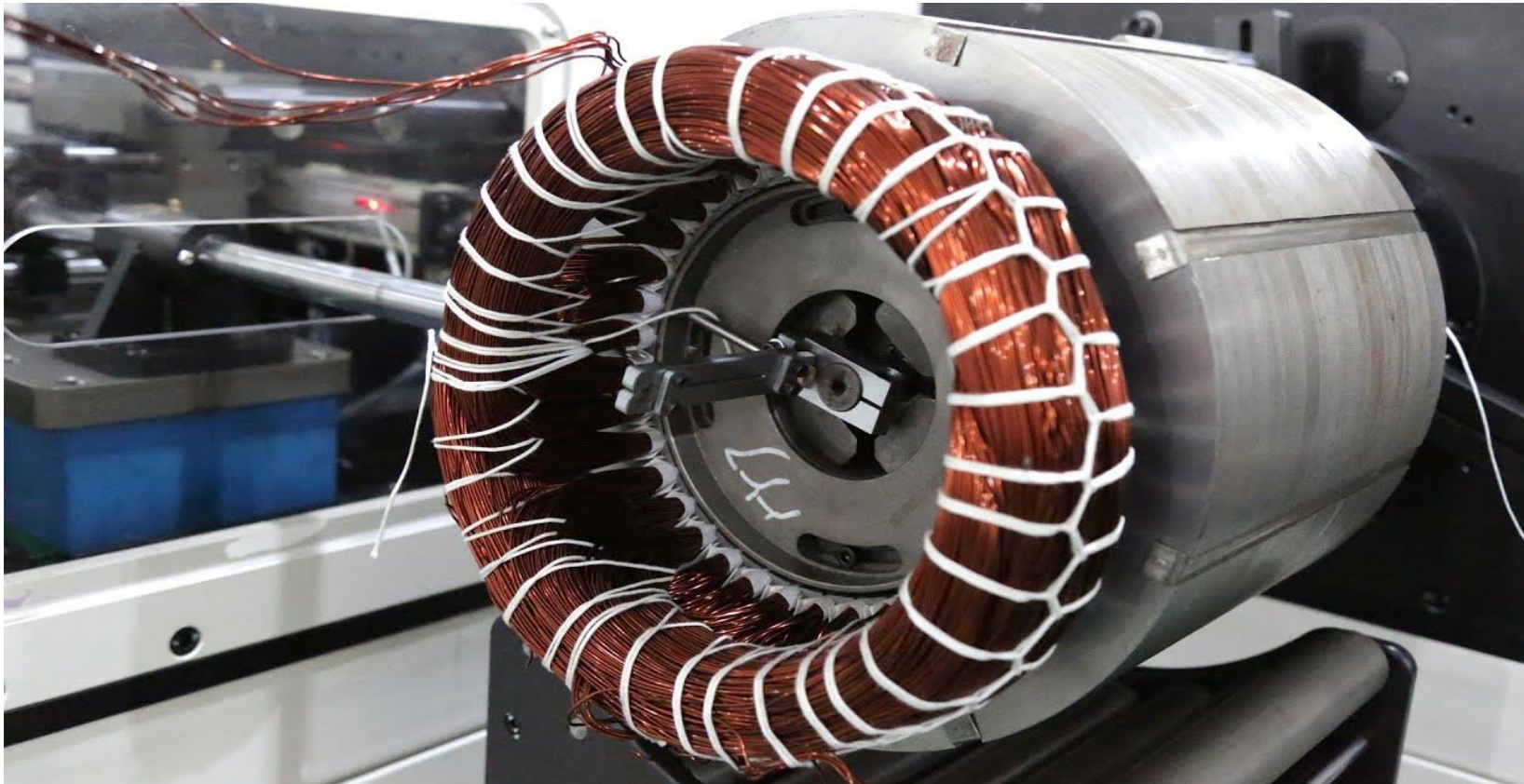




# INTRODUCTION



AC motors winding are connected in star/delta connection depending on requirement and application.



Three Phase Induction Motor Winding





# INTRODUCTION



Star and Delta connections are used in starting of three phase induction motors using STAR-DELTA Starter.



Star Delta Starter for Three phase Induction Motor



# INTRODUCTION



Delta-Star Starters are installed in cement industries for high inertial load applications.



Cement Industry





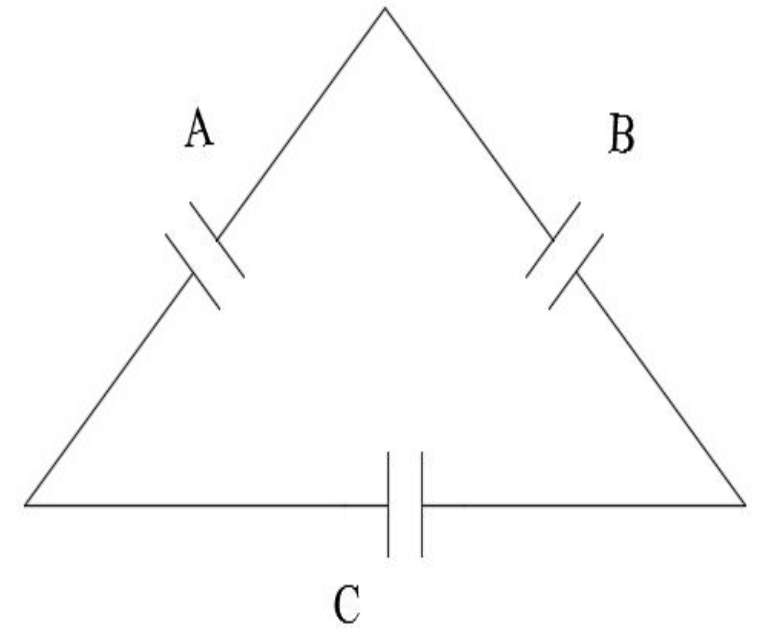
# INTRODUCTION



Power capacitors in 3 phase capacitor bank connections are either delta connected or star (wye) connected.



Delta connected capacitor bank



*Delta connection of capacitors*





# INTRODUCTION



The application of such connection is also used in high voltage direct current (HVDC) systems.



Generating Transformer



# INTRODUCTION



The application of such connection is also used in Wheatstone bridge resistance measurement device.



Wheatstone Bridge



# INTRODUCTION



STAR/Delta transformations and equivalent circuit calculations help in simplification and understanding of complex electrical circuits.



Complex Electric Circuits





# INTRODUCTION



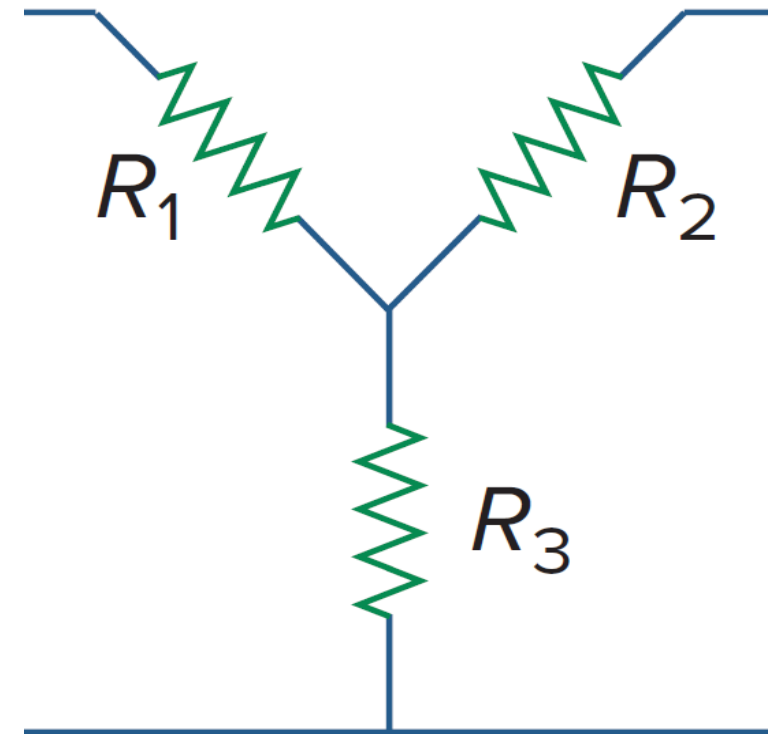
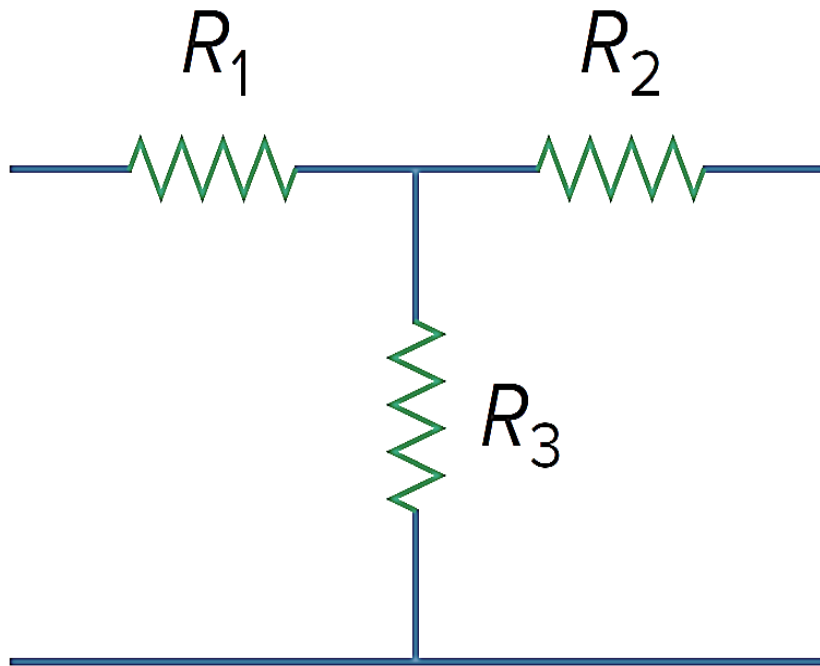
- Star/Delta connection is an arrangement of passive elements R, L and C such that the formed shape resembles a star or a delta symbol.
- These connection are neither series and nor parallel.
- Such connections are simplified using star-to-delta or delta-to-star conversion.
- Such connections are found in complex DC circuits, full bridge rectifiers.
- Such connections has larger application in three phase AC system.



# STAR CONNECTION

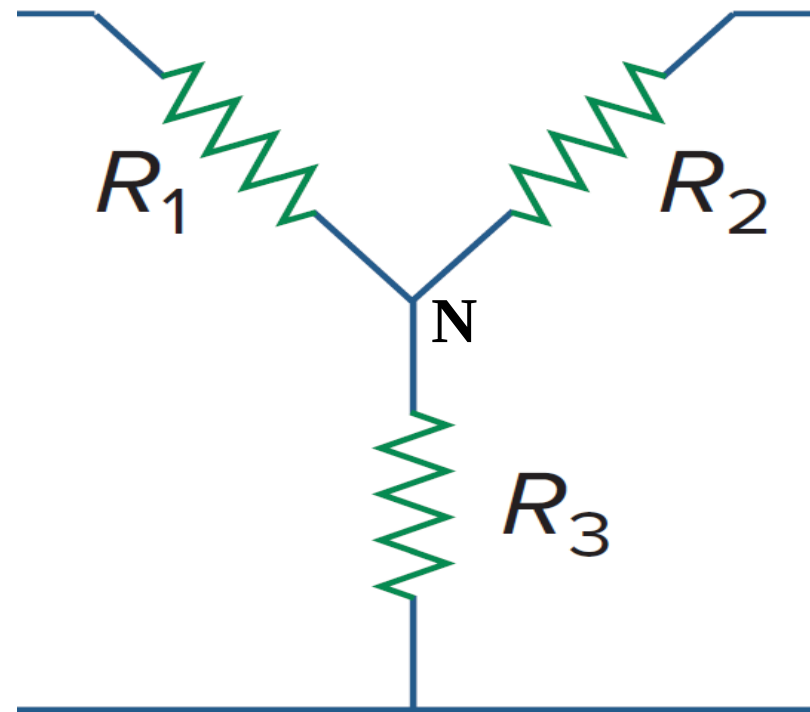


A star network is rearranged form of Tee (T) network.



# **STAR/WYE (Y) CONNECTION**

Three ends of resistors are connected in wye (Y) or star fashion. A common node point of star connection is known as neutral.



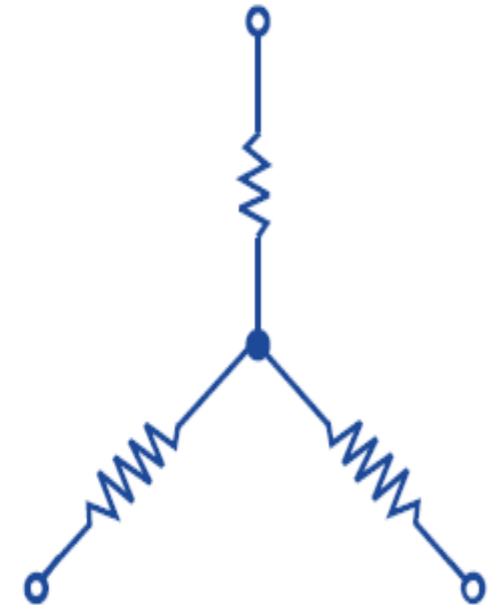
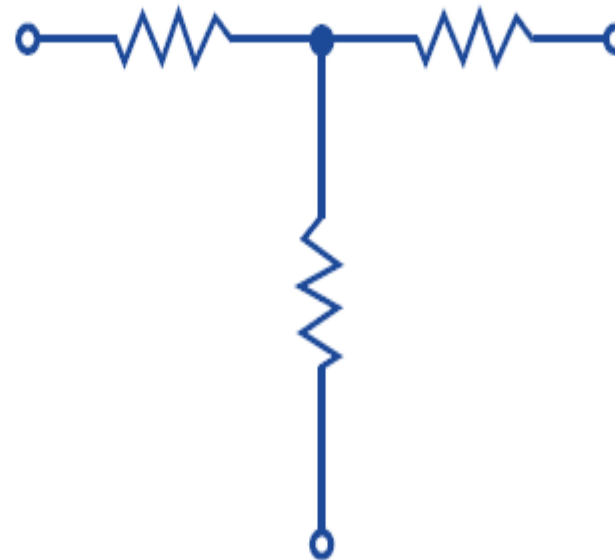
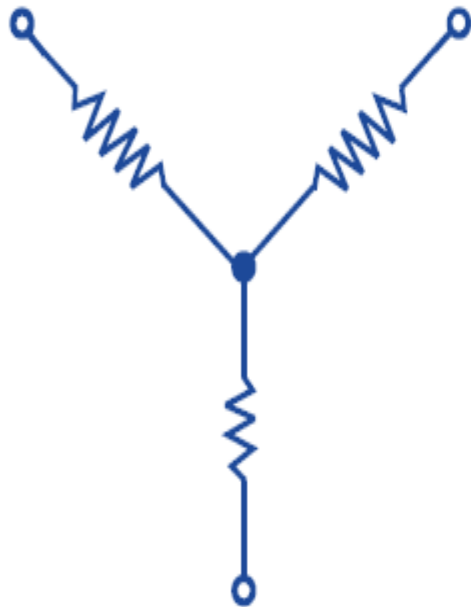




# STAR CONNECTION



Three ways in which star connection may appear in a circuit.



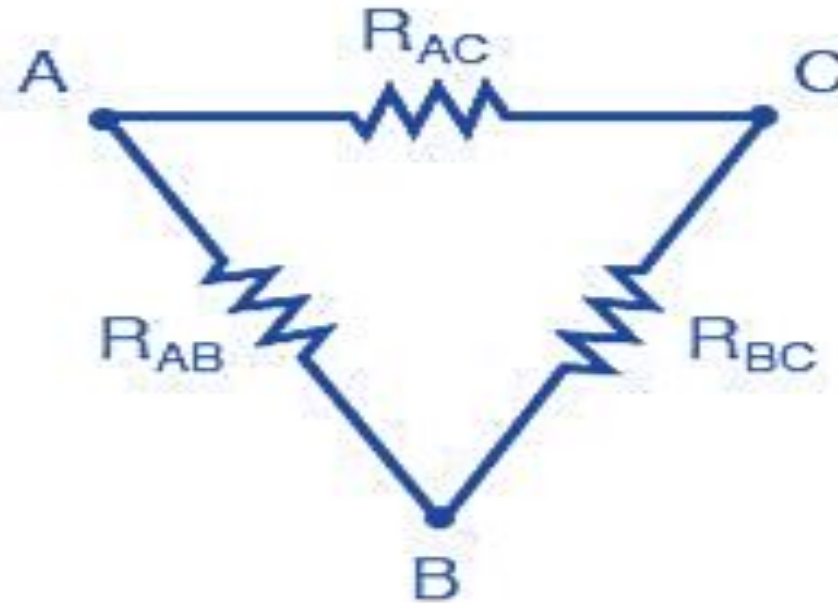


# DELTA CONNECTION



When three resistors are connected in a fashion to form a closed mesh  $\Delta$ , connection formed is known as Delta Connection.

Delta ( $\Delta$ ) network

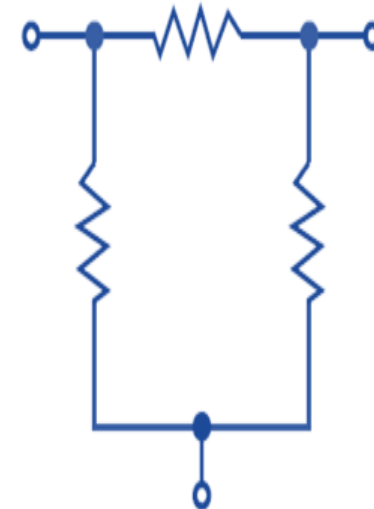
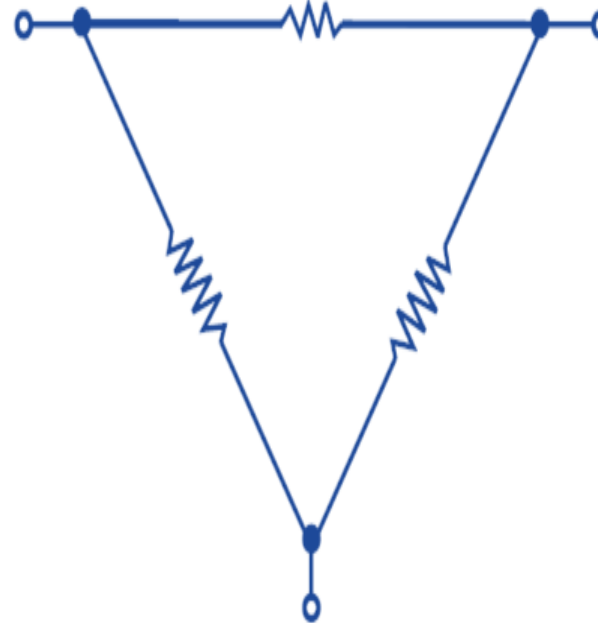
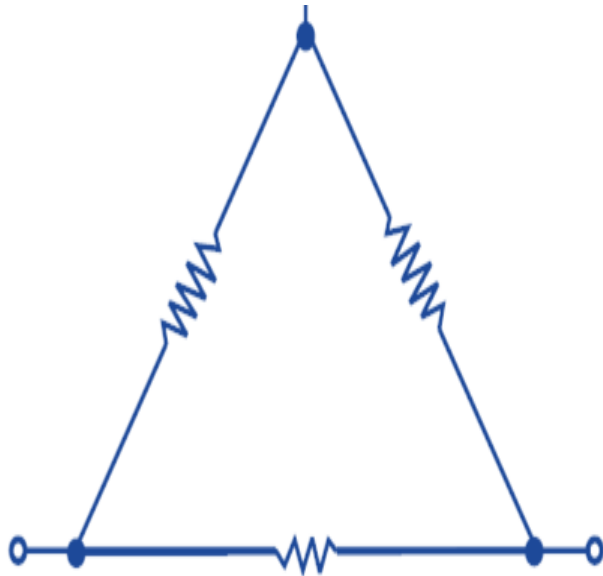




# DELTA CONNECTION



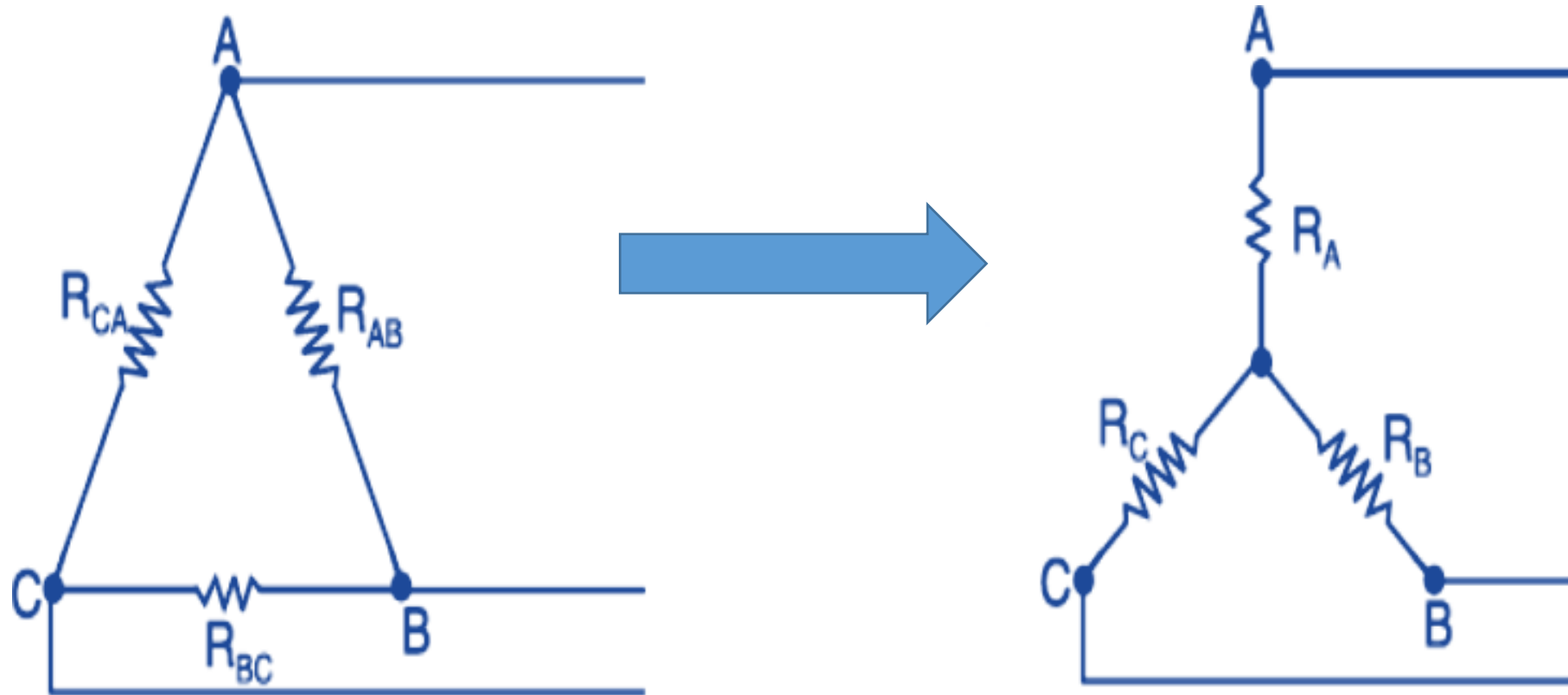
Three ways in which delta connection may appear in a circuit.





# $\Delta$ $\Delta$ $\Delta$ **DELTA TO STAR TRANSFORMATION** $\Delta$ $\Delta$ $\Delta$

Three resistors  $R_{AB}$ ,  $R_{BC}$  and  $R_{CA}$  connected in delta form and its equivalent star connection is shown below.



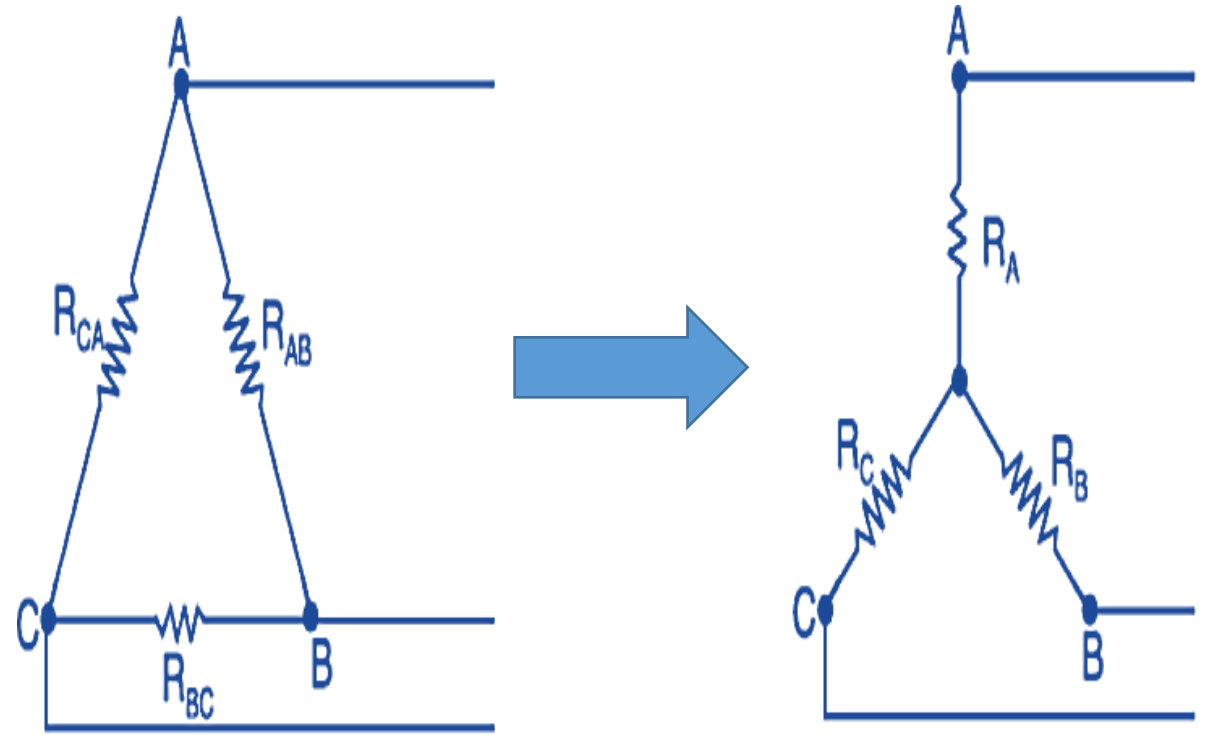
Delta and its equivalent Star

# $\Delta$ $\Delta$ $\Delta$ $\Delta$ $\Delta$ **DELTA TO STAR TRANSFORMATION** $\Delta$ $\Delta$ $\Delta$ $\Delta$ $\Delta$

- Two arrangements shown are electrically equivalent.
- Resistance between A and B for star = Resistance between A and B for delta.
- Therefore,

$$R_A + R_B = R_{AB} \parallel (R_{BC} + R_{CA}) \quad (1)$$

$$R_A + R_B = \frac{R_{AB}(R_{BC} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}} \quad (2)$$





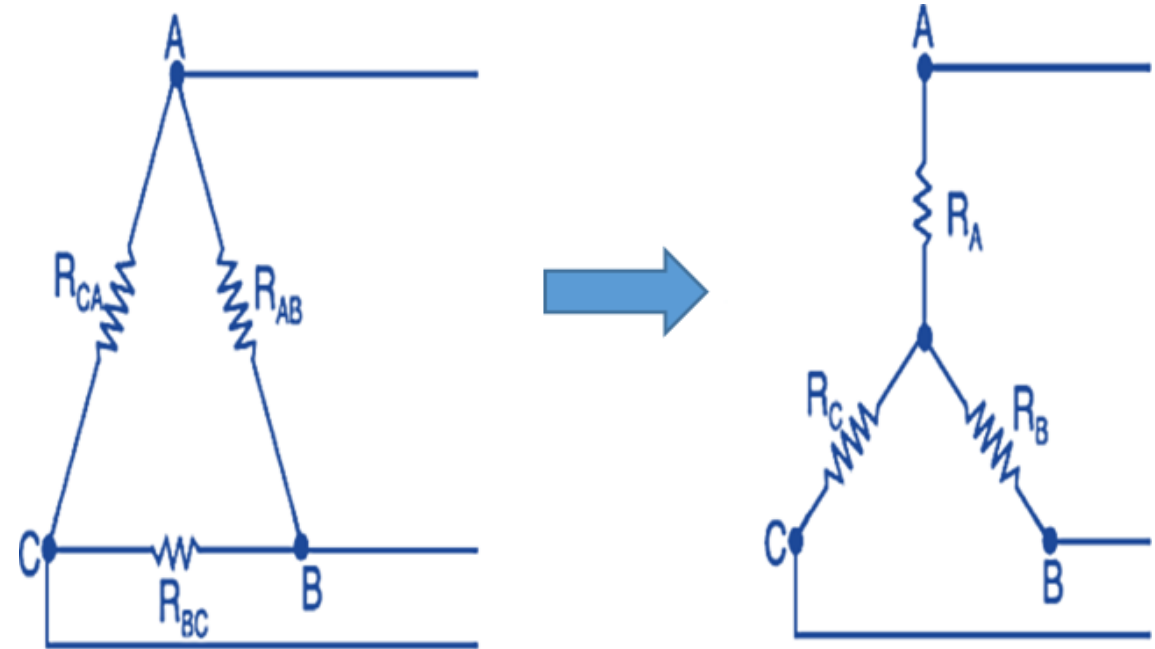
# DELTA TO STAR TRANSFORMATION



- Similarly for resistance between two terminals B-C and C-A,

$$\Rightarrow R_B + R_C = \frac{R_{BC} (R_{CA} + R_{AB})}{R_{AB} + R_{BC} + R_{CA}} \quad (3)$$

$$\Rightarrow R_C + R_A = \frac{R_{CA} (R_{AB} + R_{BC})}{R_{AB} + R_{BC} + R_{CA}} \quad (4)$$





# DELTA TO STAR TRANSFORMATION

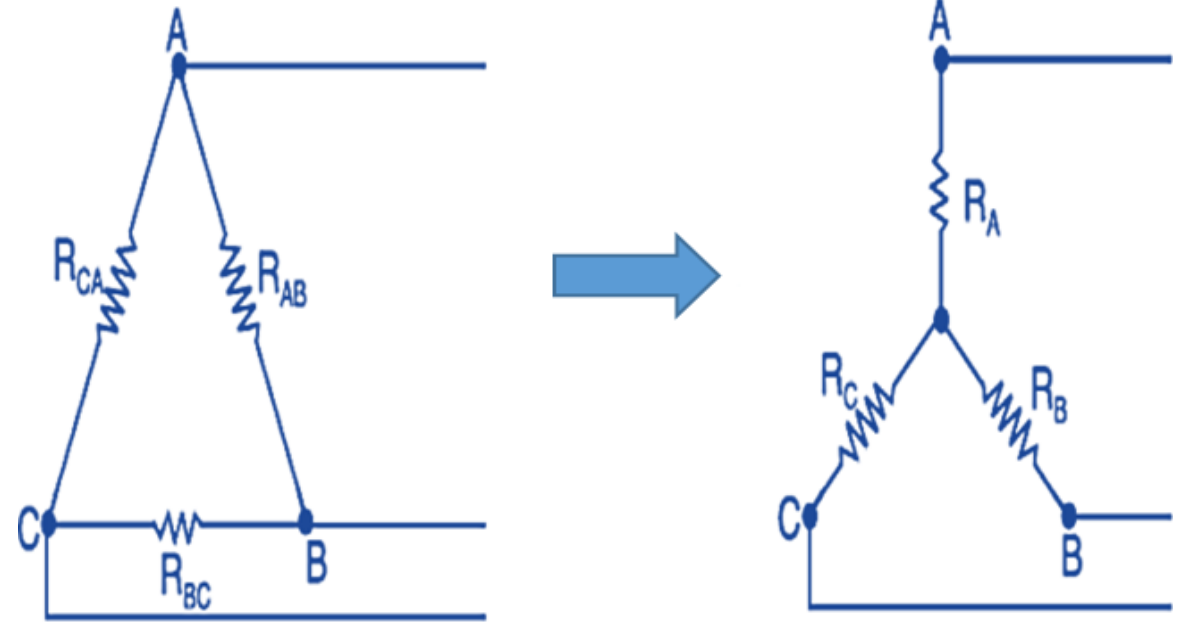


- The objective is to find  $R_A, R_B$  and  $R_C$  in terms of  $R_{AB}, R_{BC}$  and  $R_{CA}$ .
- Subtracting (3) from (2) and adding to (4) we obtain,

$$\Rightarrow R_A = \frac{R_{AB} R_{CA}}{R_{AB} + R_{BC} + R_{CA}} \quad (5)$$

$$\Rightarrow R_B = \frac{R_{BC} R_{AB}}{R_{AB} + R_{BC} + R_{CA}} \quad (6)$$

$$\Rightarrow R_C = \frac{R_{CA} R_{BC}}{R_{AB} + R_{BC} + R_{CA}} \quad (7)$$





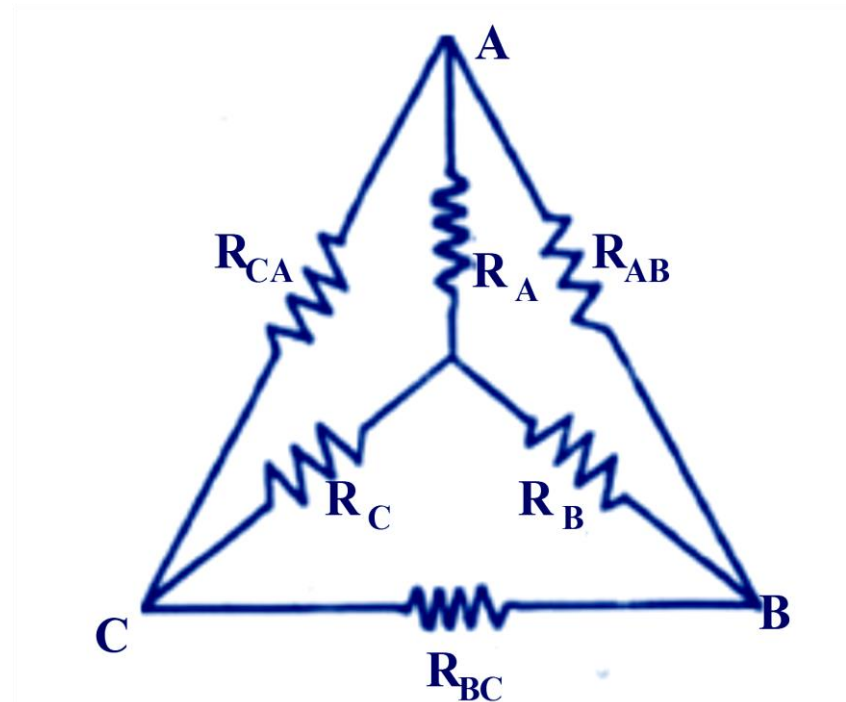


# DELTA TO STAR TRANSFORMATION



- Easy way to remember delta to star transformation is,

$$\text{Any arm of star connection} = \frac{\text{Product of two adjacent arms of } \Delta}{\text{Sum of arms of } \Delta}$$

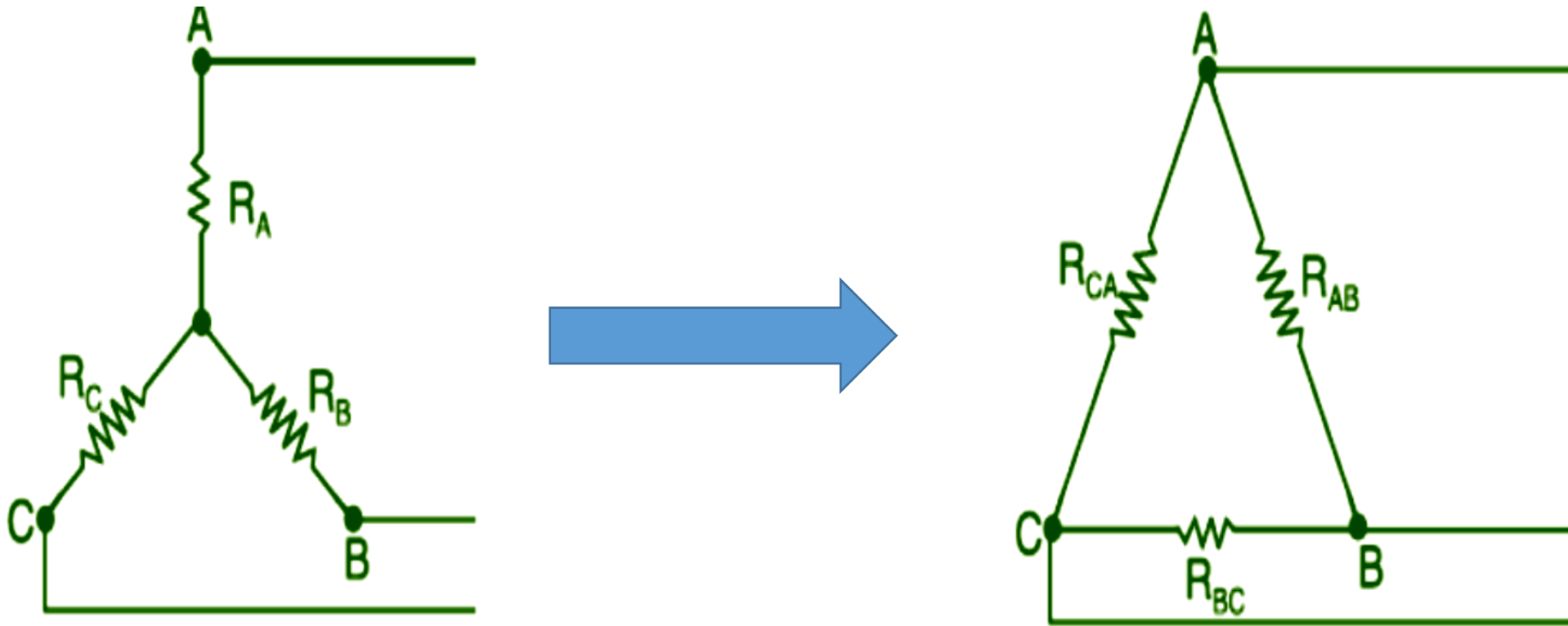




# STAR TO DELTA TRANSFORMATION



Three resistors  $R_A, R_B$  and  $R_C$  connected in star formation and its equivalent delta connection is shown below



Star and its Equivalent Delta



# STAR TO DELTA TRANSFORMATION



- Dividing (5) by (6) we obtain,

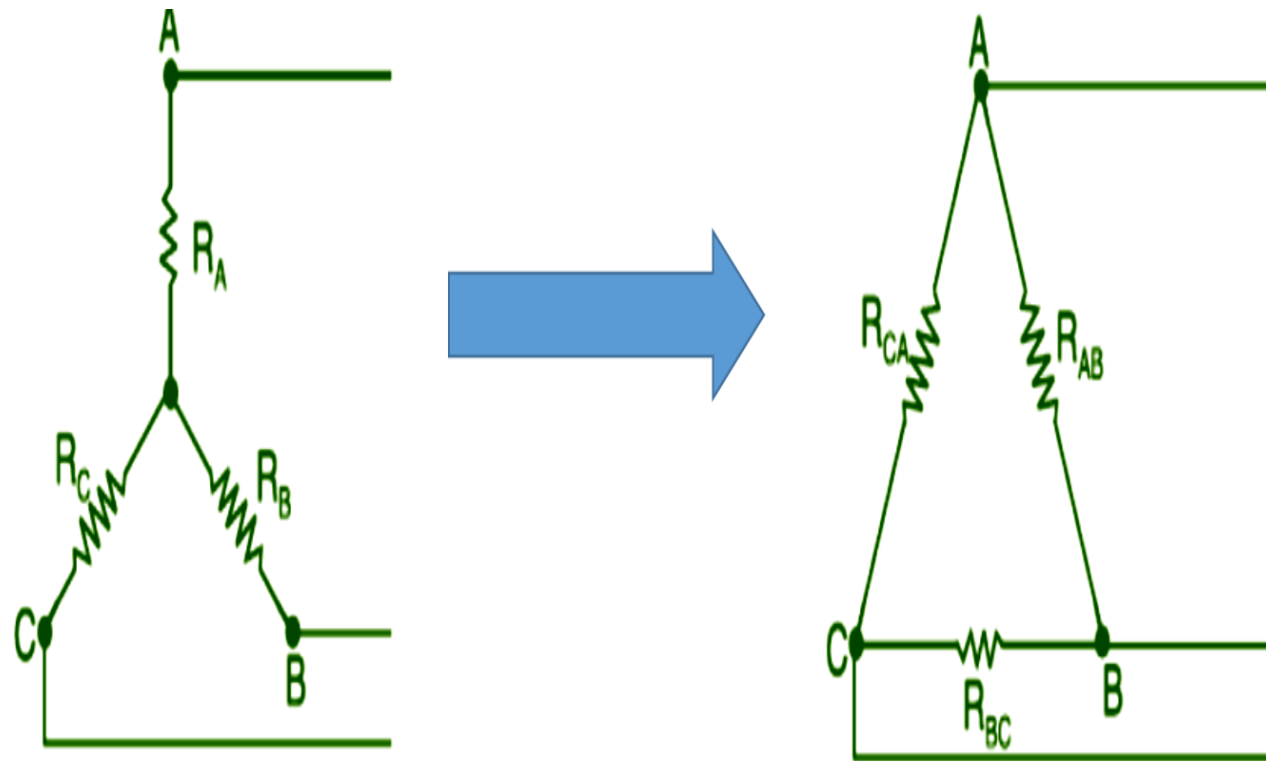
$$\frac{R_A}{R_B} = \frac{R_{CA}}{R_{BC}} \quad (8)$$

$$\Rightarrow R_{CA} = \frac{R_A R_{BC}}{R_B} \quad (9)$$

- Dividing (5) by (7) we obtain,

$$\frac{R_A}{R_C} = \frac{R_{AB}}{R_{BC}} \quad (10)$$

$$\Rightarrow R_{AB} = \frac{R_A R_{BC}}{R_C} \quad (11)$$





# STAR TO DELTA TRANSFORMATION



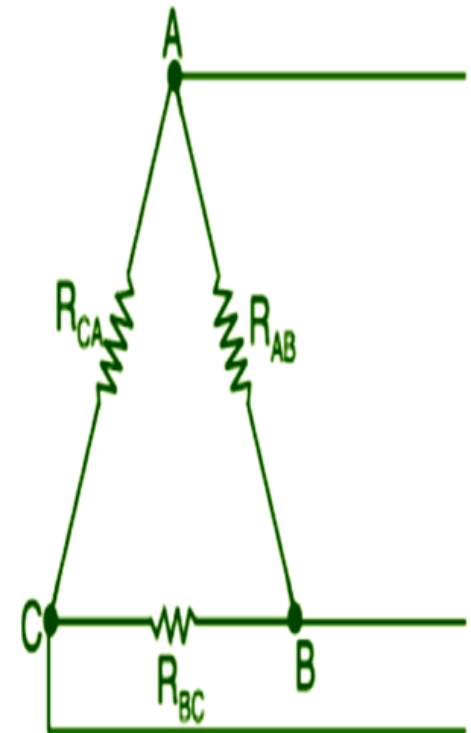
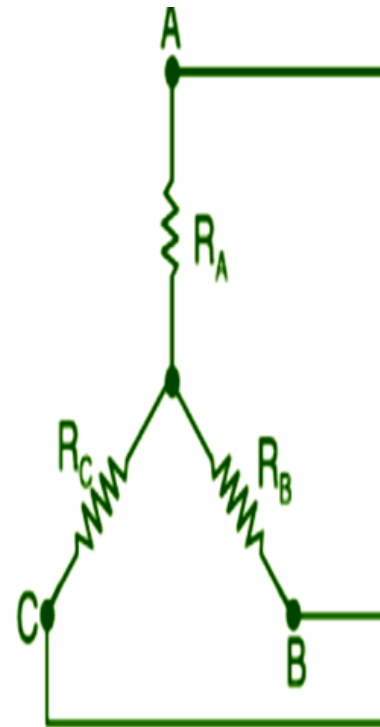
- Substituting (9) and (11) into (5),

$$R_{BC} = R_B + R_C + \frac{R_B R_C}{R_A} \quad (12)$$

- Similarly,

$$R_{AB} = R_A + R_B + \frac{R_A R_B}{R_C} \quad (13)$$

$$R_{CA} = R_C + R_A + \frac{R_C R_A}{R_B} \quad (14)$$



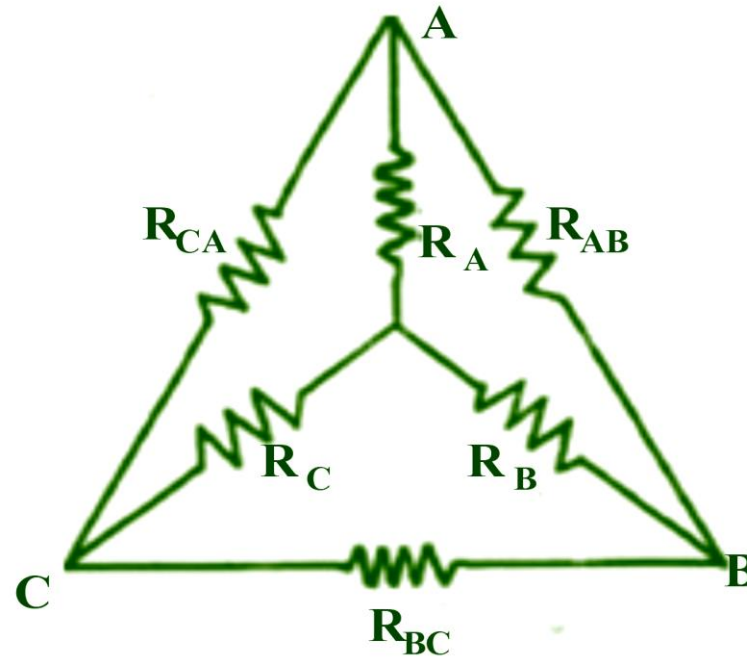




# STAR TO DELTA TRANSFORMATION



- Easy way to remember star to delta transformation is,  
**Resistance between two terminals of  $\Delta$  =**  
**Sum of star resistances connected to those terminals +**  
**product of same two resistances divided by the third**





# STAR\DELTA TRANSFORMATION



- If a star network has all resistances equal to  $R$ , its equivalent delta has all resistances equal to ?
- If a delta network has all resistances equal to  $R$ , its equivalent star has all resistances equal to ?



# STAR\DELTA TRANSFORMATION



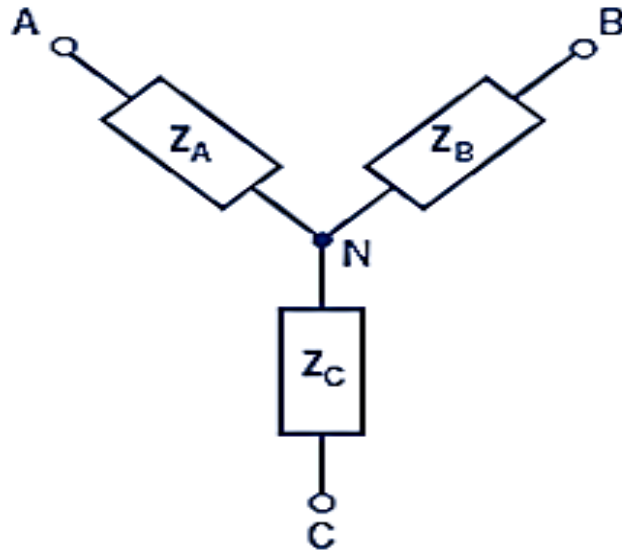
- If a star network has all resistances equal to  $R$ , its equivalent delta has all resistances equal to  $3R$ .
- If a delta network has all resistances equal to  $R$ , its equivalent star has all resistances equal to  $R/3$ .

# SUMMARY

## CIRCUIT

## CONVERSION FORMULAS (UNBALANCED)

## CONVERSION FORMULAS (BALANCED)



$$z_A = \frac{z_1 z_3}{z_1 + z_2 + z_3}$$

$$z_B = \frac{z_2 z_3}{z_1 + z_2 + z_3}$$

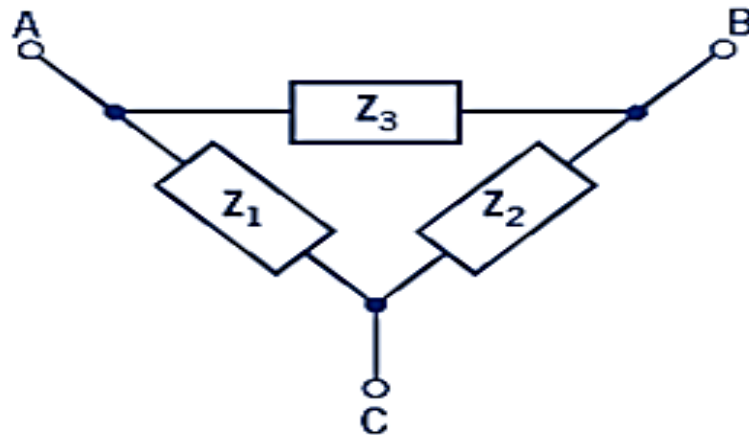
$$z_C = \frac{z_1 z_2}{z_1 + z_2 + z_3}$$

When

$$z_1 = z_2 = z_3 = z$$

then

$$z_A = z_B = z_C = \frac{z}{3}$$



$$z_1 = \frac{z_A + z_C + \frac{z_A z_C}{z_B}}{z_B}$$

$$z_2 = \frac{z_B + z_C + \frac{z_B z_C}{z_A}}{z_A}$$

$$z_3 = \frac{z_A + z_B + \frac{z_A z_B}{z_C}}{z_C}$$

When

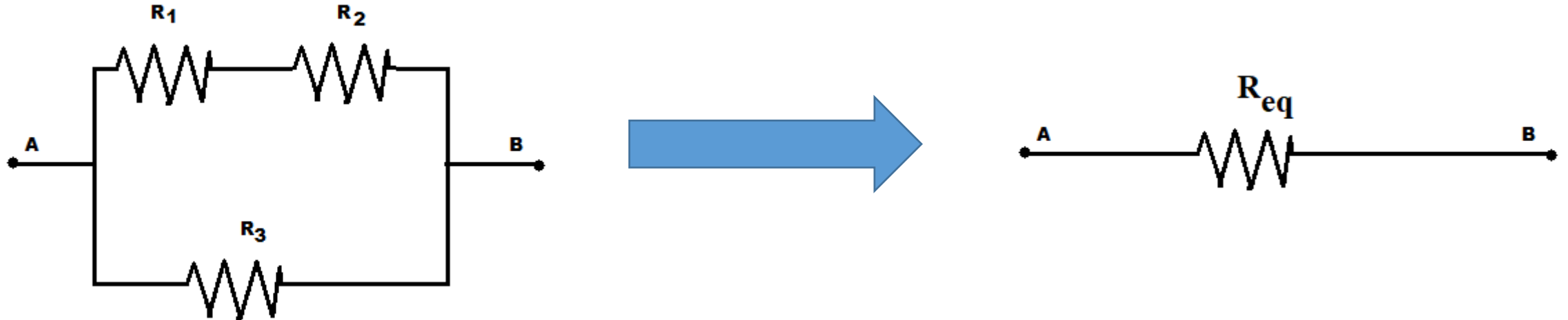
$$z_A = z_B = z_C = z_Y$$

then

$$z_1 = z_2 = z_3 = 3z_Y$$

# EQUIVALENT RESISTANCE

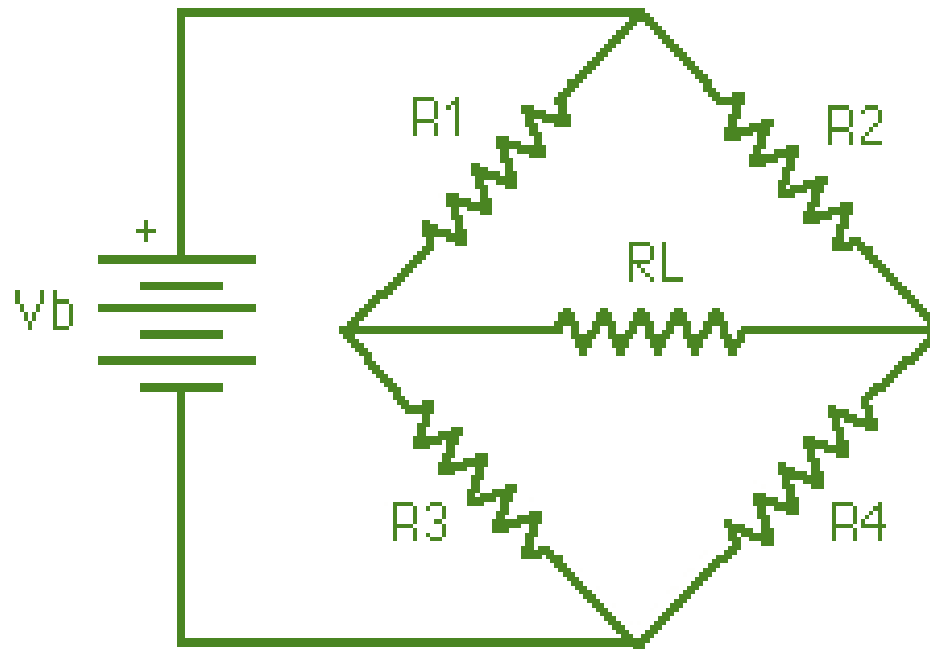
- The equivalent resistance of a circuit or network between its any two points (or terminals) is that single resistance which can replace the entire circuit between these points (or terminals).





## DEFINITIONS

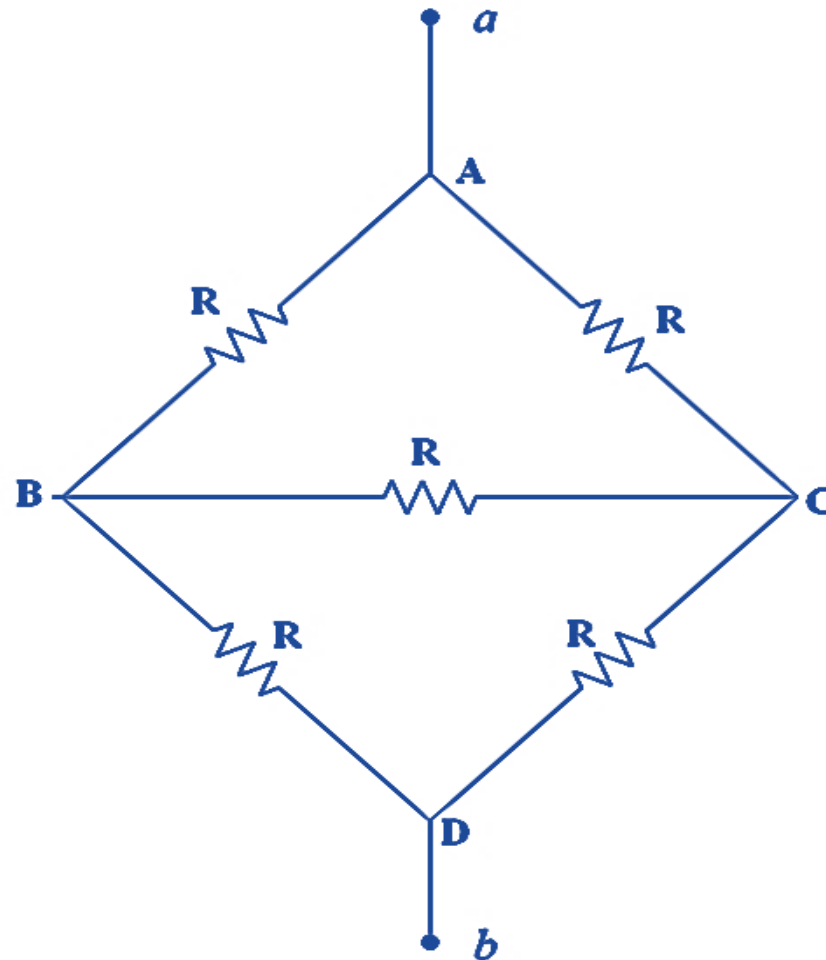
- **STAR/DELTA CIRCUITS:** These circuits generally possess star/delta configurations and needs to be simplified using necessary transformations and are converted into series parallel circuits.



Neither Series Nor Parallel Circuit

## EQUIVALENT RESISTANCE OF STAR/DELTA CIRCUIT

- The circuit is a combination of neither series nor parallel circuits.





## EQUIVALENT RESISTANCE OF STAR/DELTA CIRCUIT



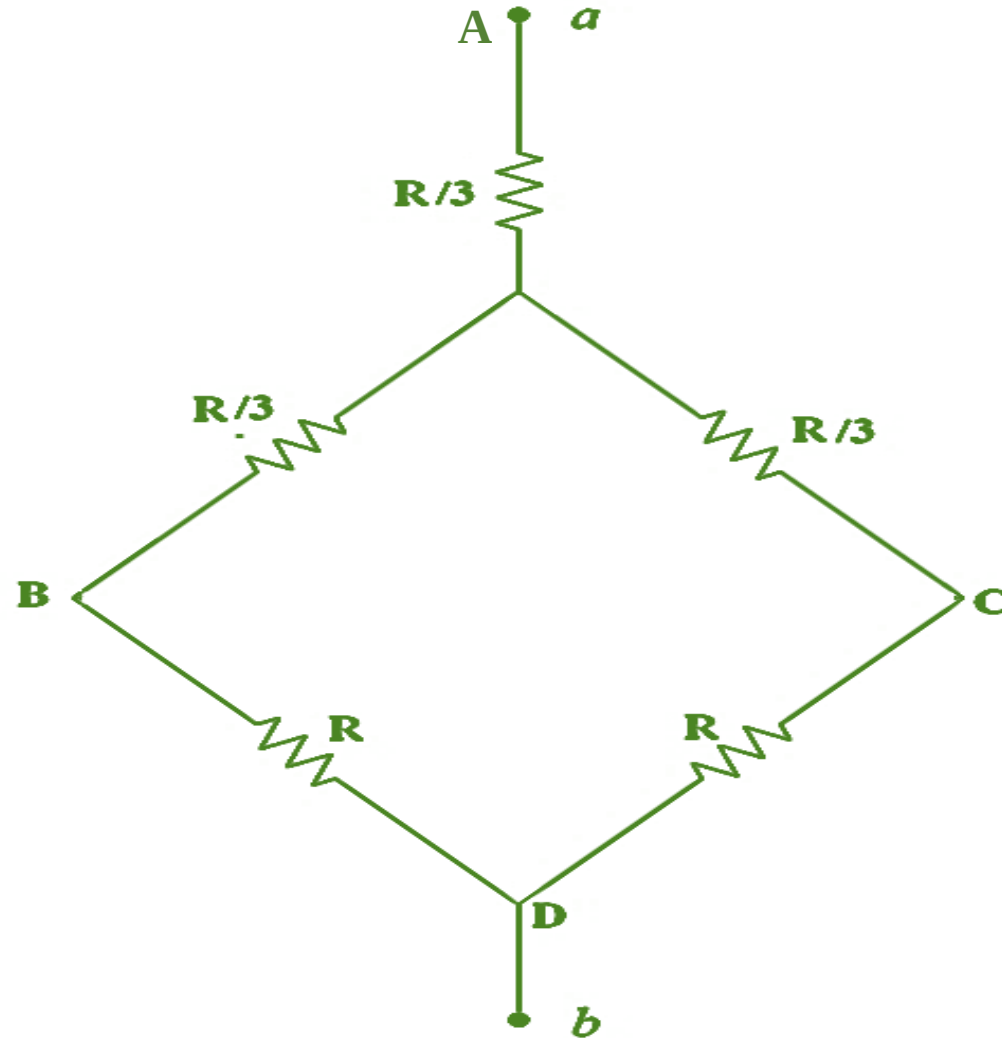
- RULE: Such circuit form star/delta. Use star delta transformation and convert to equivalent series-parallel circuit.
- Changing Delta formed by points A,B,C into equivalent Star,

$$R_Y = \frac{R \times R}{R + R + R} = R / 3$$

- The circuit formed is now combination of series-parallel circuit.
- The series-parallel circuit is further simplified to series circuit.

## EQUIVALENT RESISTANCE OF STAR DELTA CIRCUIT

The circuit now becomes a simple series-parallel circuit and can be solved easily.

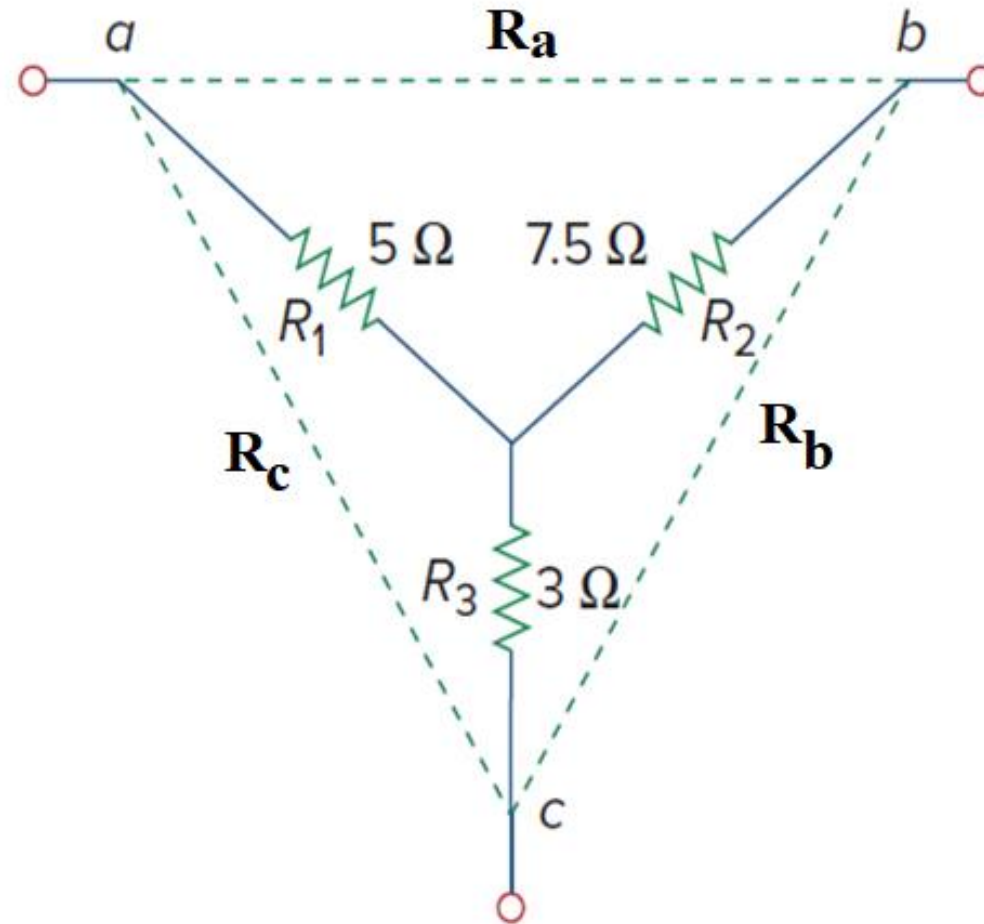




# EXERCISE/NUMERICAL ANALYSIS



Q. Convert the Y network to an equivalent  $\Delta$  network.



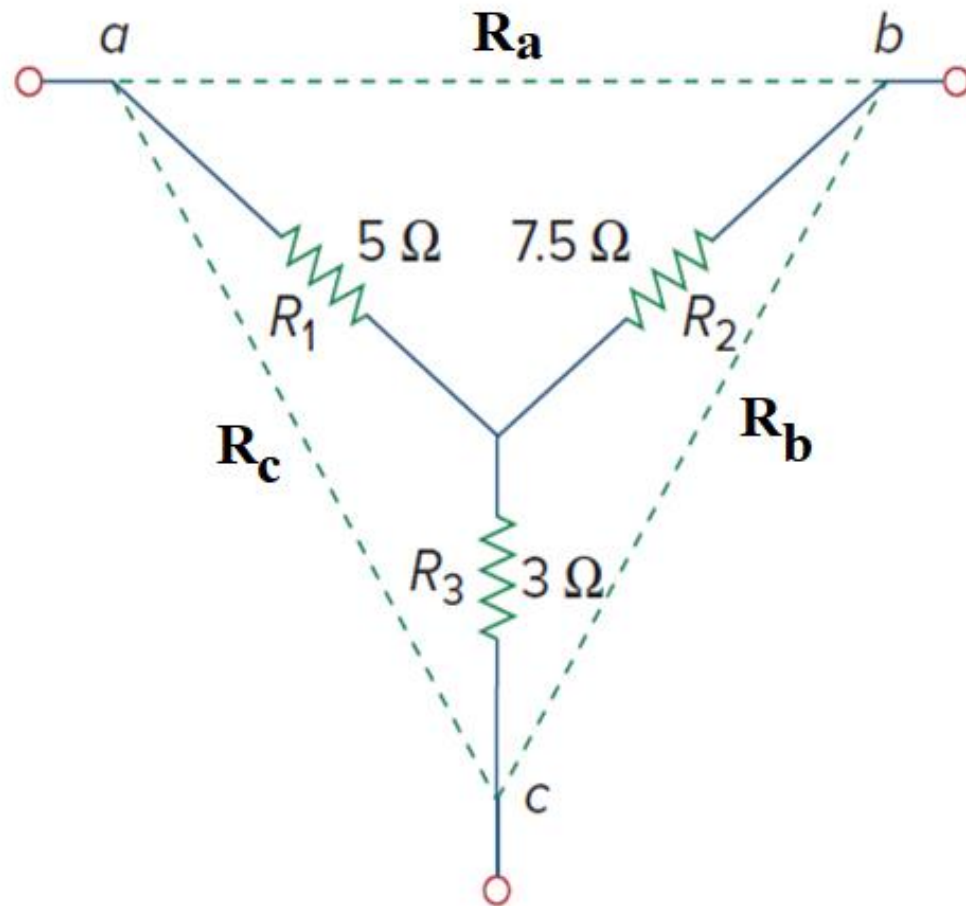




# EXERCISE/NUMERICAL ANALYSIS



Soln:



$$R_a = 7.5 + 5 + \frac{7.5 \times 5}{3} = 25 \text{ ohms}$$

$$R_b = 7.5 + 3 + \frac{7.5 \times 3}{5} = 15 \text{ ohms}$$

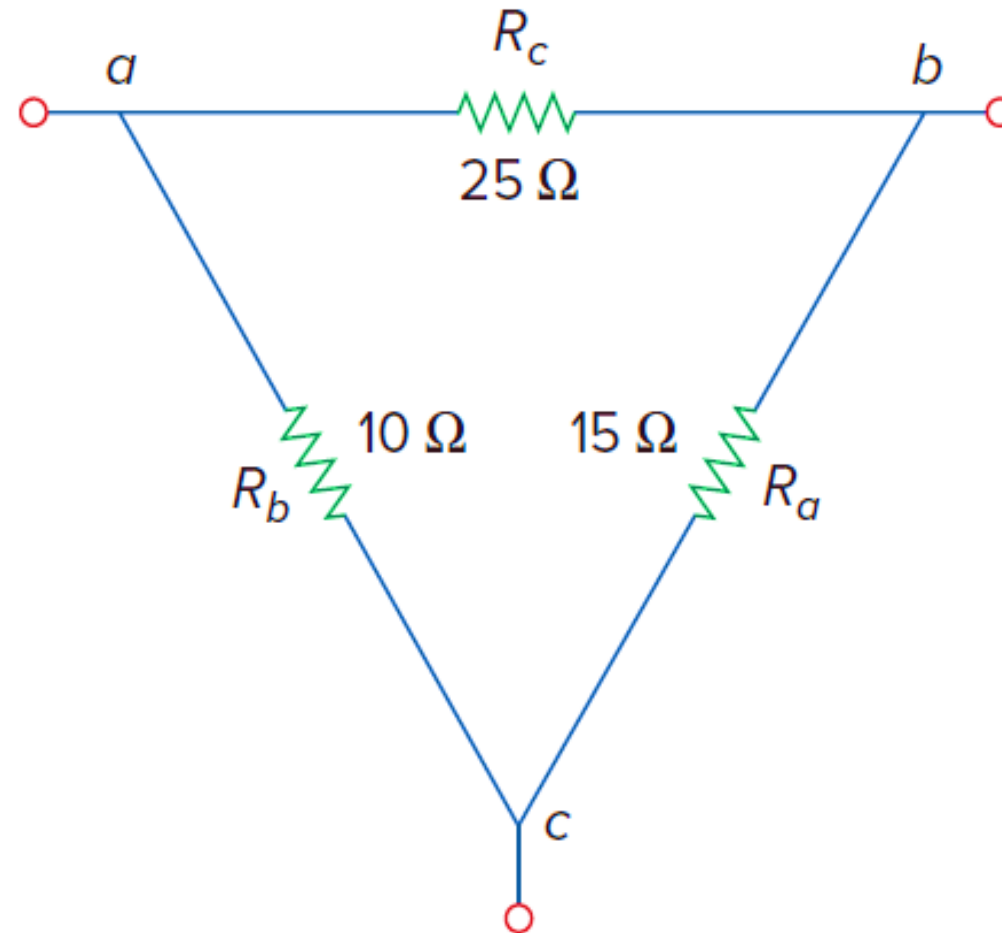
$$R_c = 5 + 3 + \frac{5 \times 3}{7.5} = 10 \text{ ohms}$$



# EXERCISE/NUMERICAL ANALYSIS



Q. Convert the  $\Delta$  network to an equivalent Y network.





# EXERCISE/NUMERICAL ANALYSIS

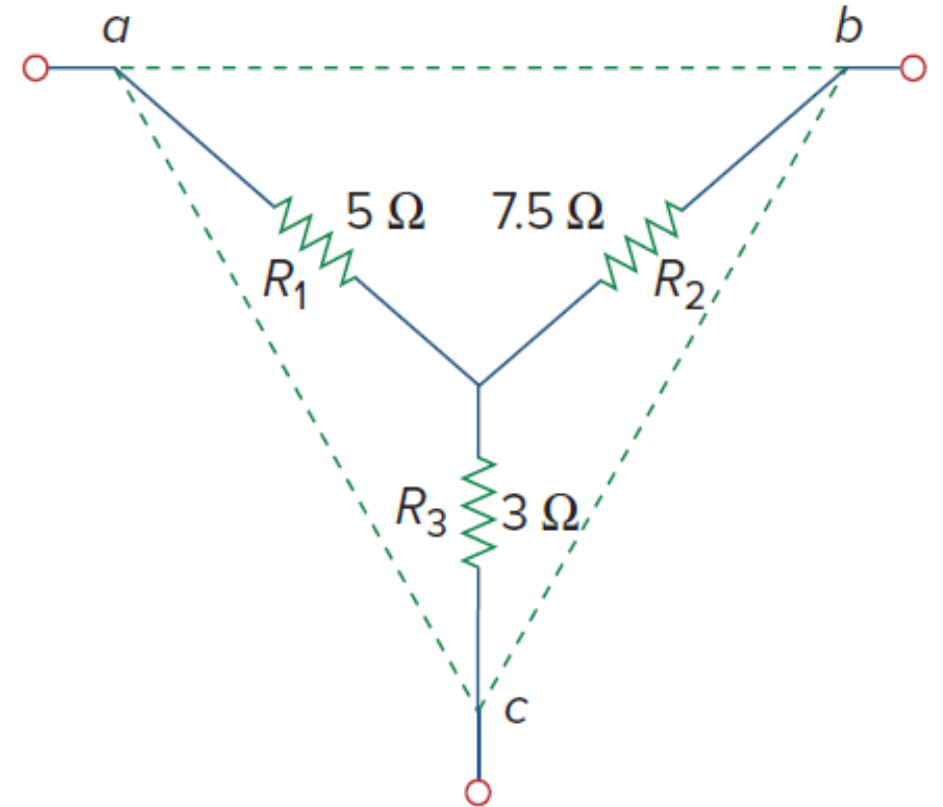


Soln:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{10 \times 25}{15 + 10 + 25} = 5 \text{ ohms}$$

$$R_2 = \frac{R_c R_a}{R_a + R_b + R_c} = \frac{25 \times 15}{50} = 7.5 \text{ ohms}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{15 \times 10}{50} = 3 \text{ ohms}$$

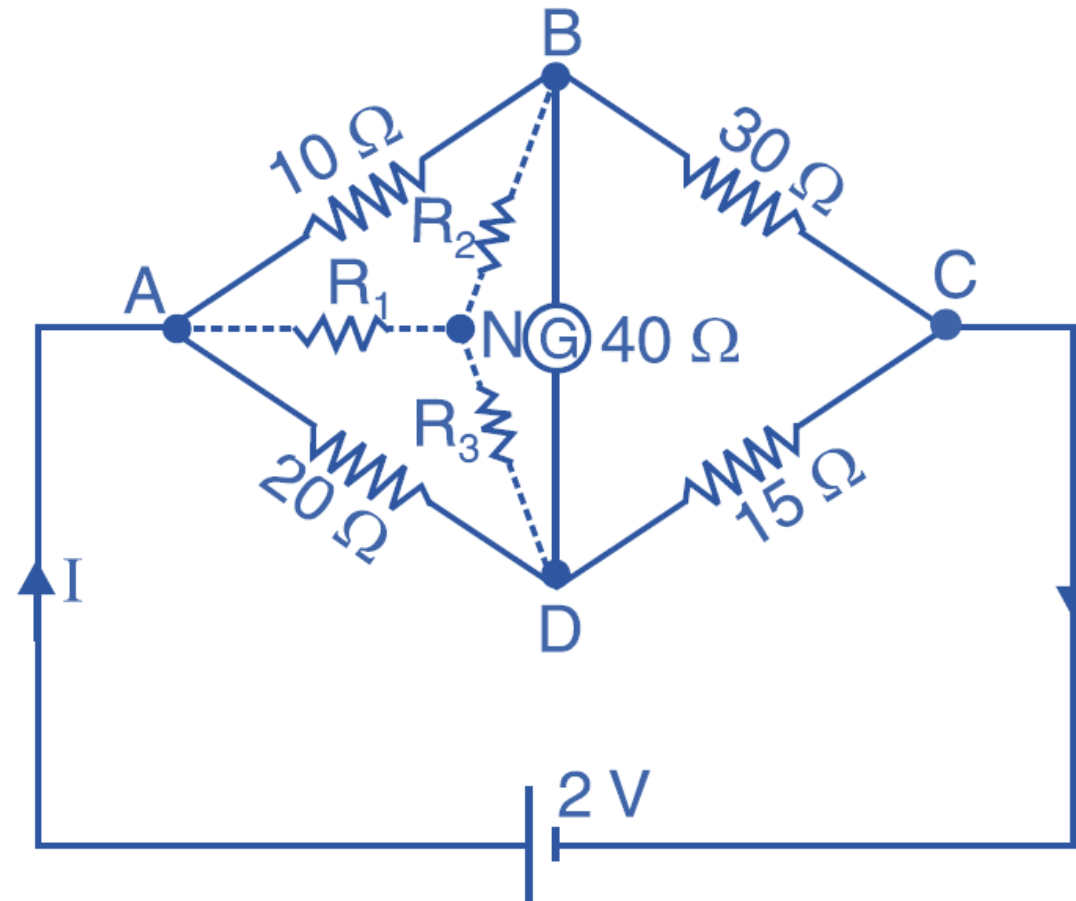




# EXERCISE/NUMERICAL ANALYSIS



Q. Using delta/star transformation, find equivalent resistance across AC.





# EXERCISE/NUMERICAL ANALYSIS

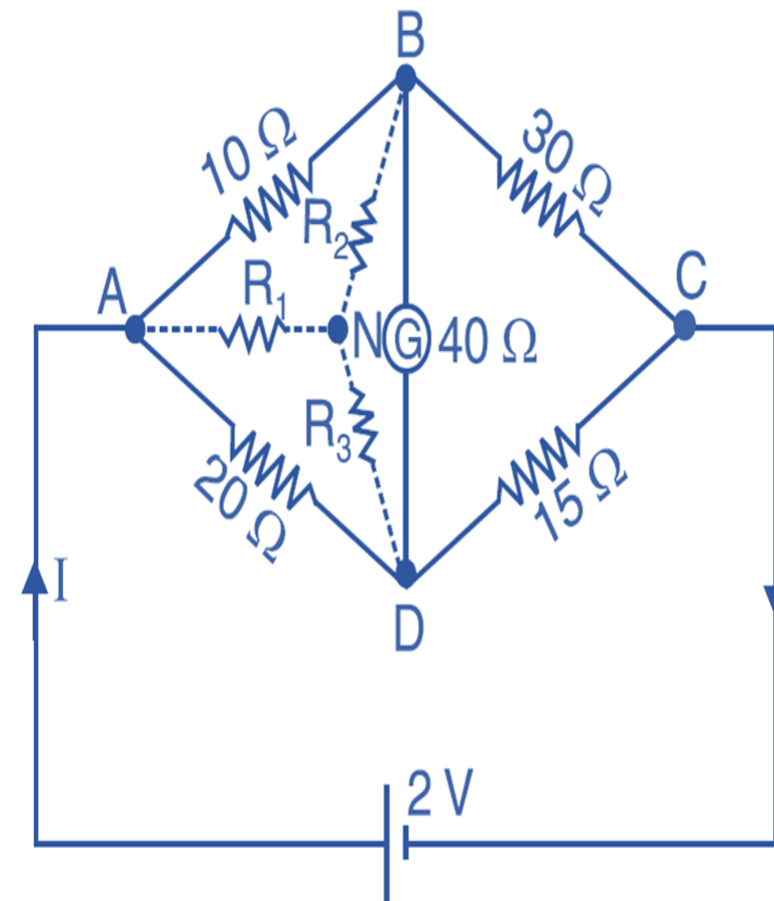


Soln: Delta can be replaced by equivalent star-connected resistances,

$$R_1 = \frac{R_{AB} R_{DA}}{R_{AB} + R_{DA} + R_{BD}} = \frac{10 \times 20}{10 + 40 + 20} = 2.86 \text{ ohms}$$

$$R_2 = \frac{R_{AB} R_{BD}}{R_{AB} + R_{DA} + R_{BD}} = \frac{10 \times 40}{10 + 40 + 20} = 5.72 \text{ ohms}$$

$$R_3 = \frac{R_{DA} R_{BD}}{R_{AB} + R_{DA} + R_{BD}} = \frac{20 \times 40}{10 + 40 + 20} = 11.4 \text{ ohms}$$

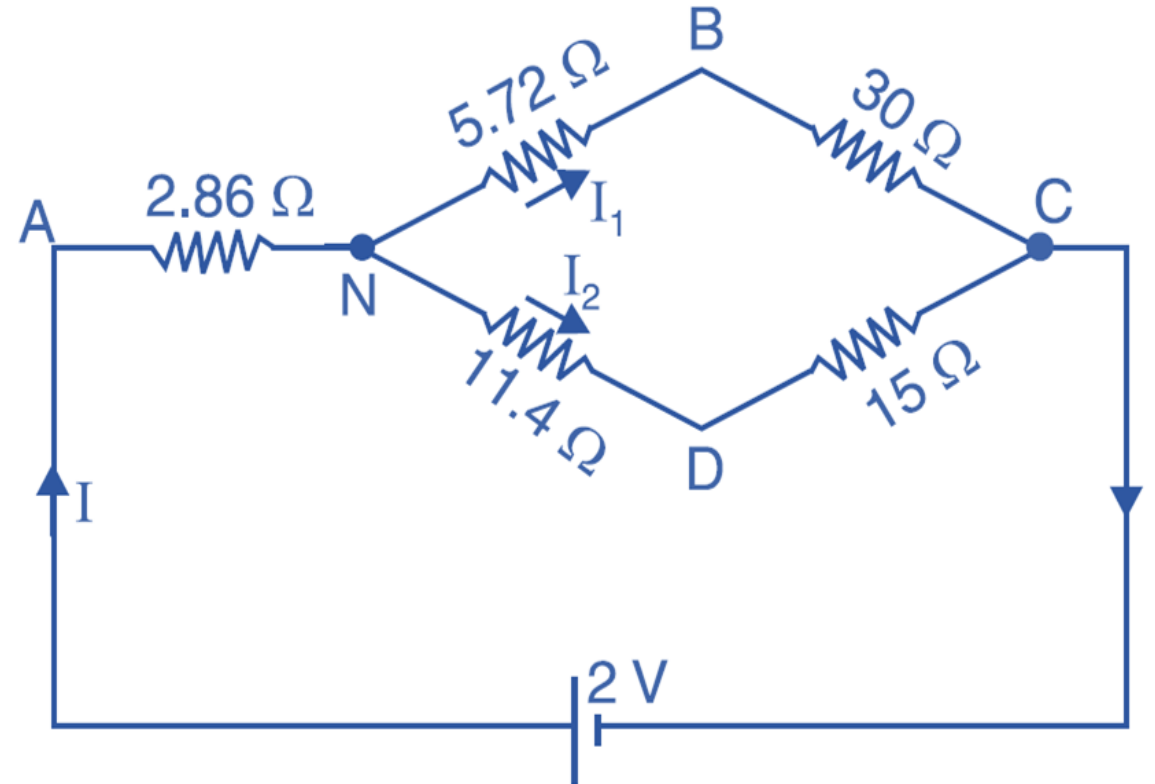
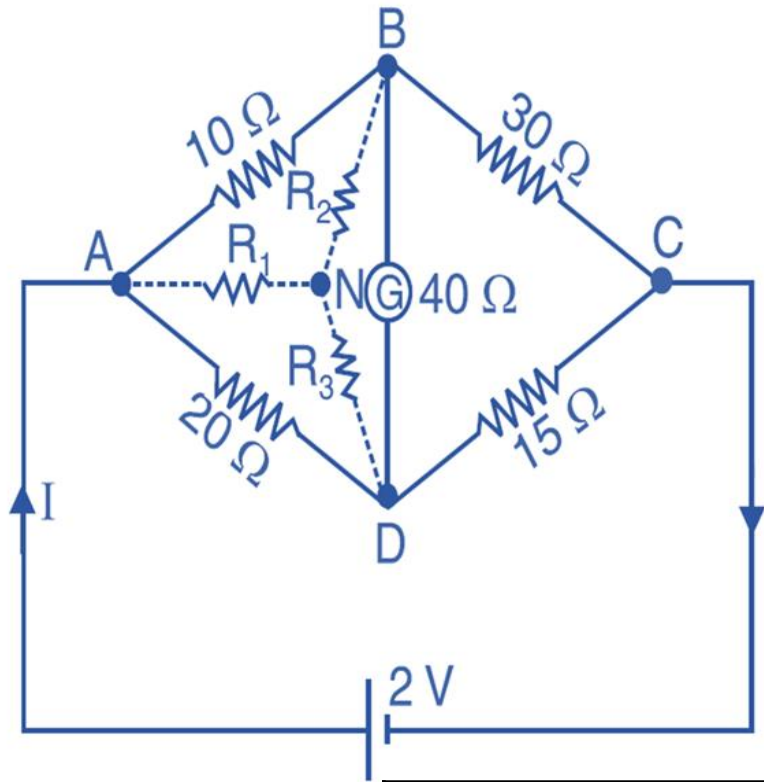




# EXERCISE/NUMERICAL ANALYSIS



Figure now becomes,



$$R_{AC} = 2.86 + \frac{(30 + 5.72) \times (15 + 11.4)}{(30 + 5.72) + (15 + 11.4)} = 18.04 \text{ ohms}$$

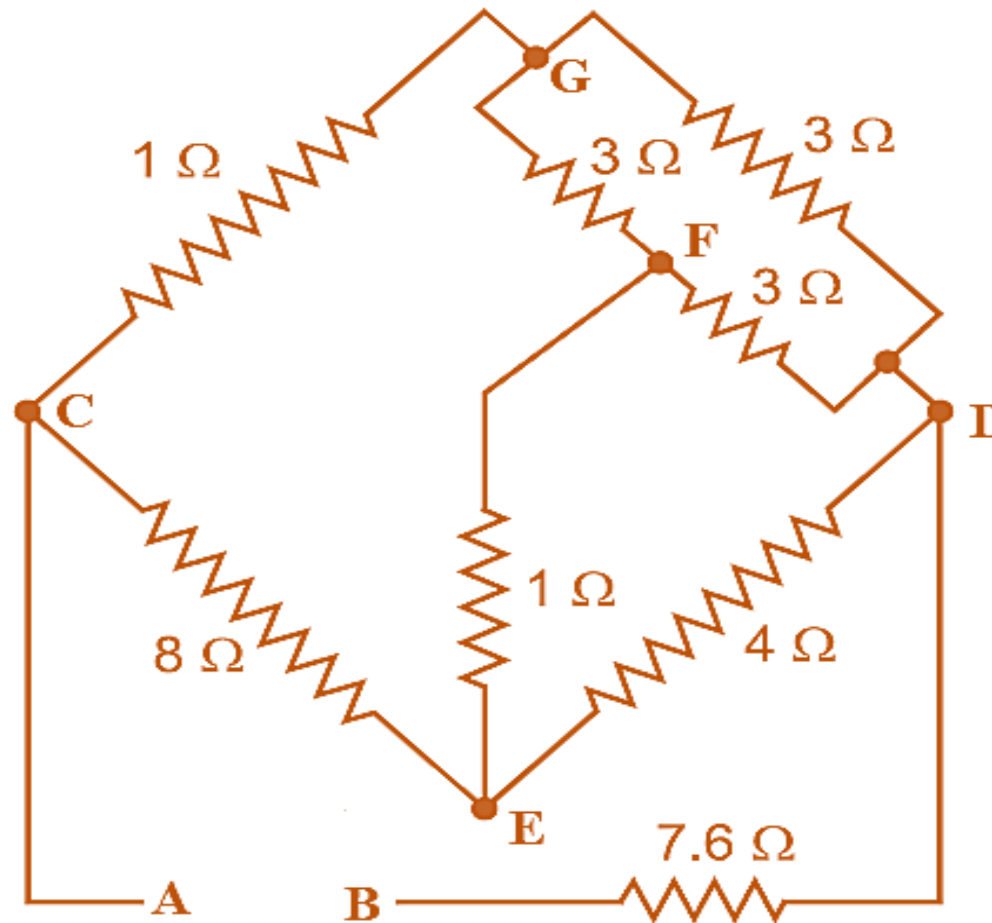




# EXERCISE/NUMERICAL ANALYSIS



Q. Calculate equivalent resistance across terminals A and B.

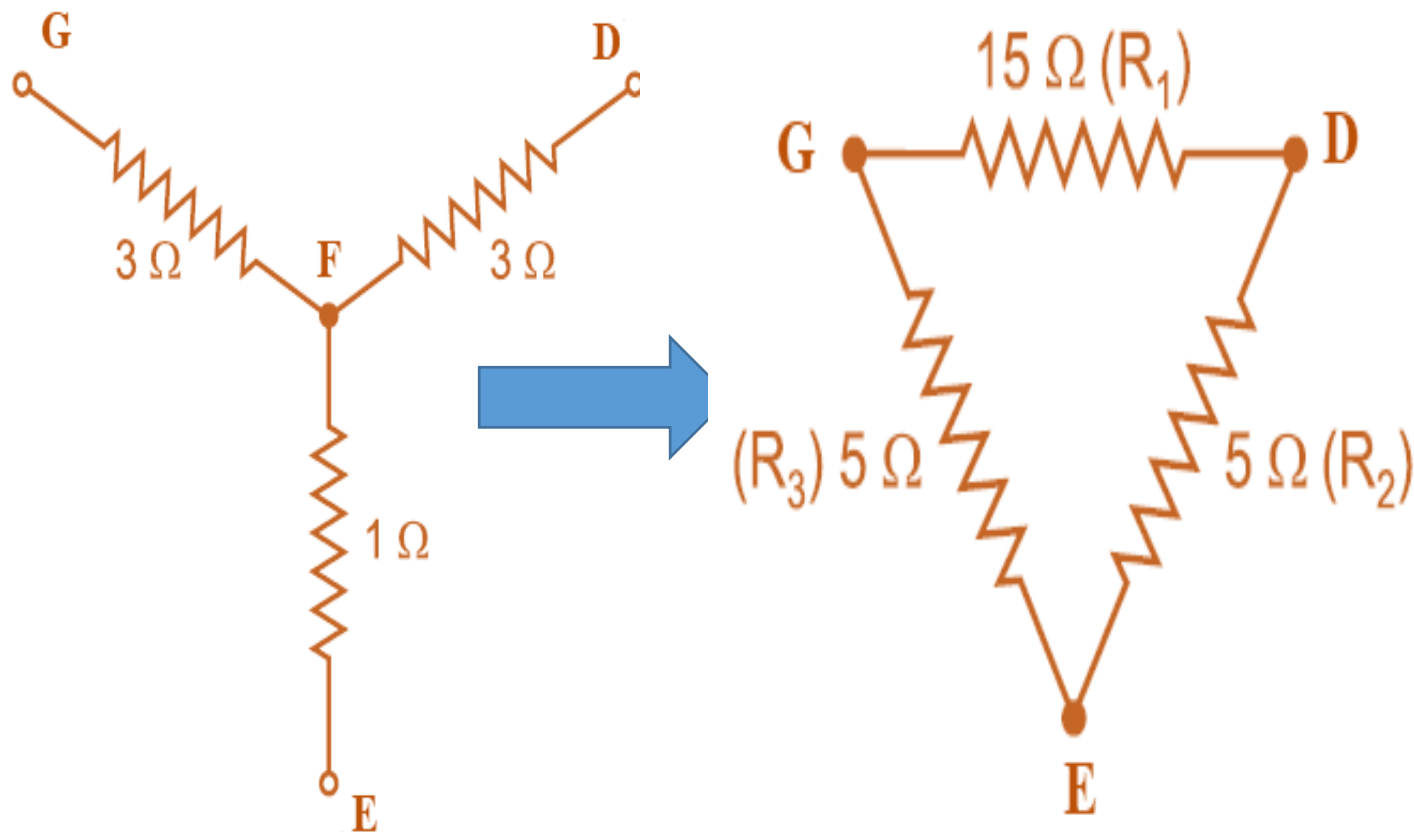




# EXERCISE/NUMERICAL ANALYSIS



Soln: Converting inner STAR (3 ohms, 3 ohms and 1 ohms ) into Delta.



$$R_1 = 3 + 3 + \frac{3 \times 3}{1} = 15 \text{ ohms}$$

$$R_2 = 3 + 1 + \frac{3 \times 1}{3} = 5 \text{ ohms}$$

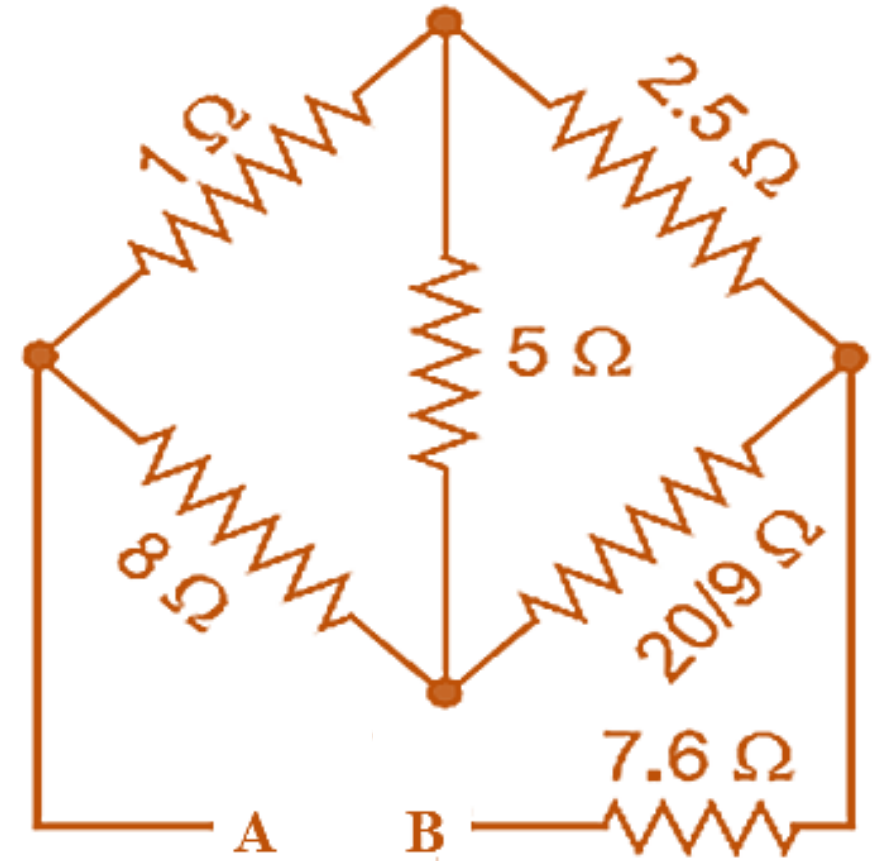
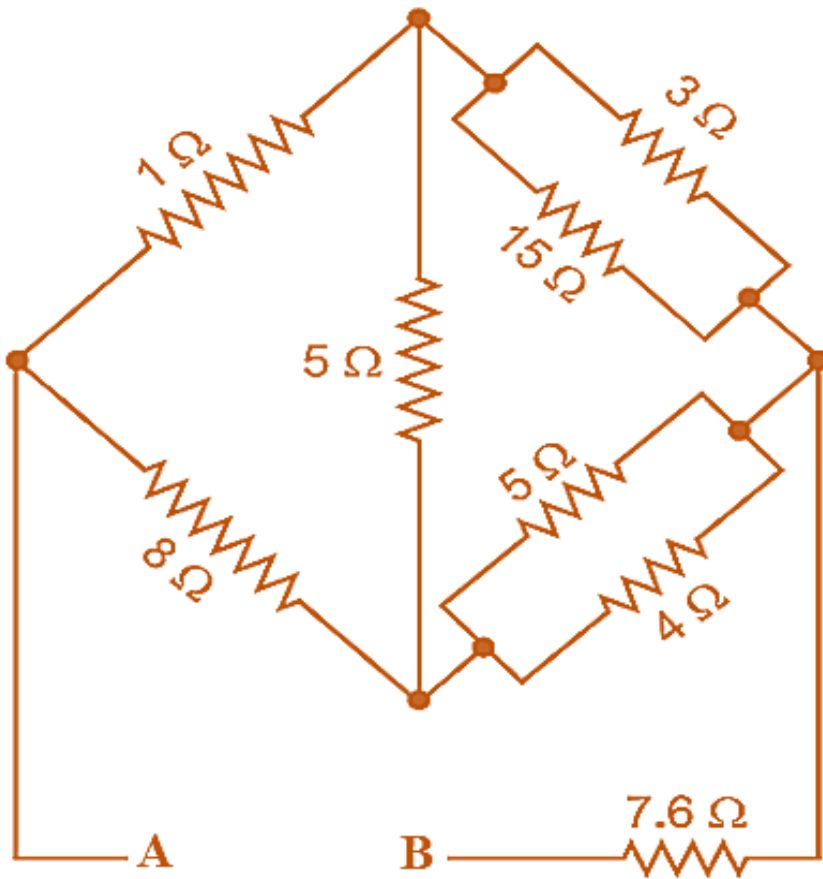
$$R_3 = 1 + 3 + \frac{1 \times 3}{3} = 5 \text{ ohms}$$



# EXERCISE/NUMERICAL ANALYSIS



Circuit now becomes,

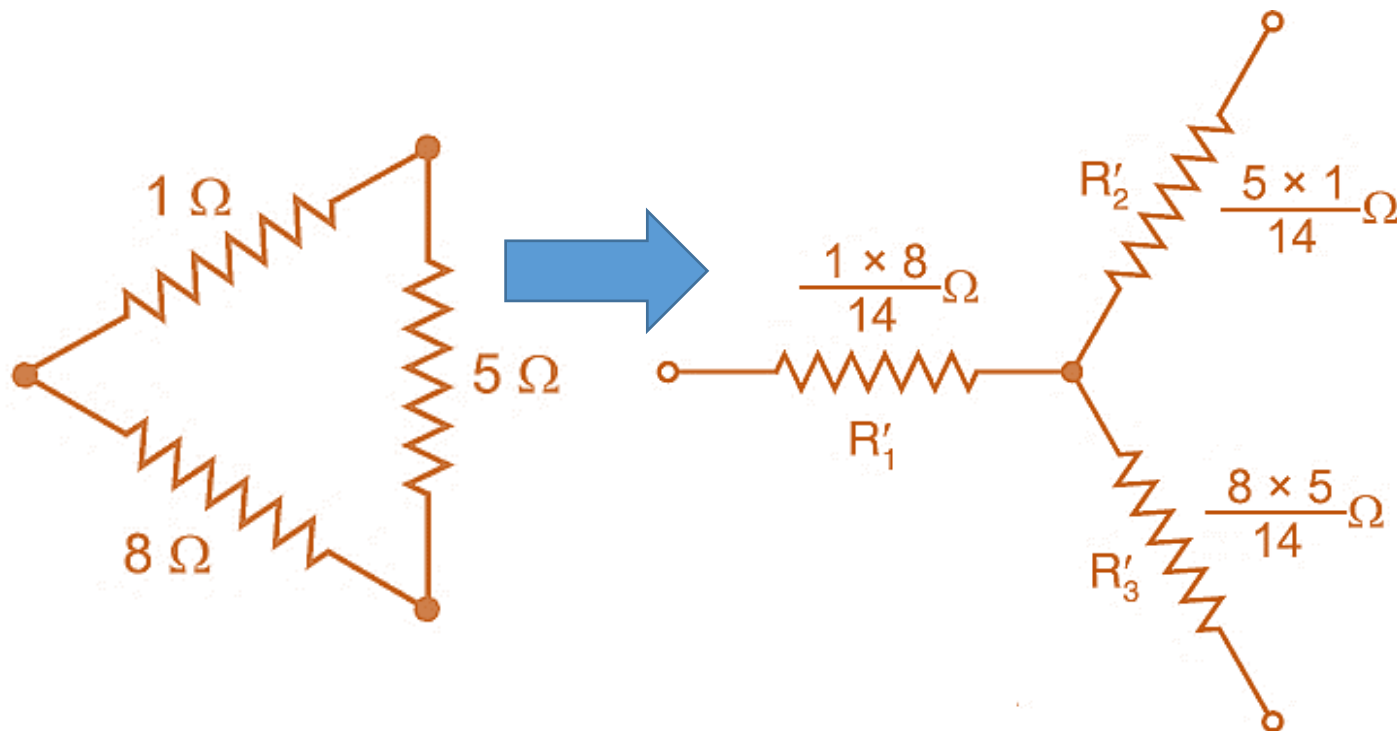




# EXERCISE/NUMERICAL ANALYSIS



Delta-connected resistances 1  $\Omega$ , 5  $\Omega$  and 8 are converted in star,



$$R_1' = \frac{1 \times 8}{1 + 5 + 8} = \frac{4}{7} \text{ ohms}$$

$$R_2' = \frac{5 \times 1}{1 + 5 + 8} = \frac{5}{14} \text{ ohms}$$

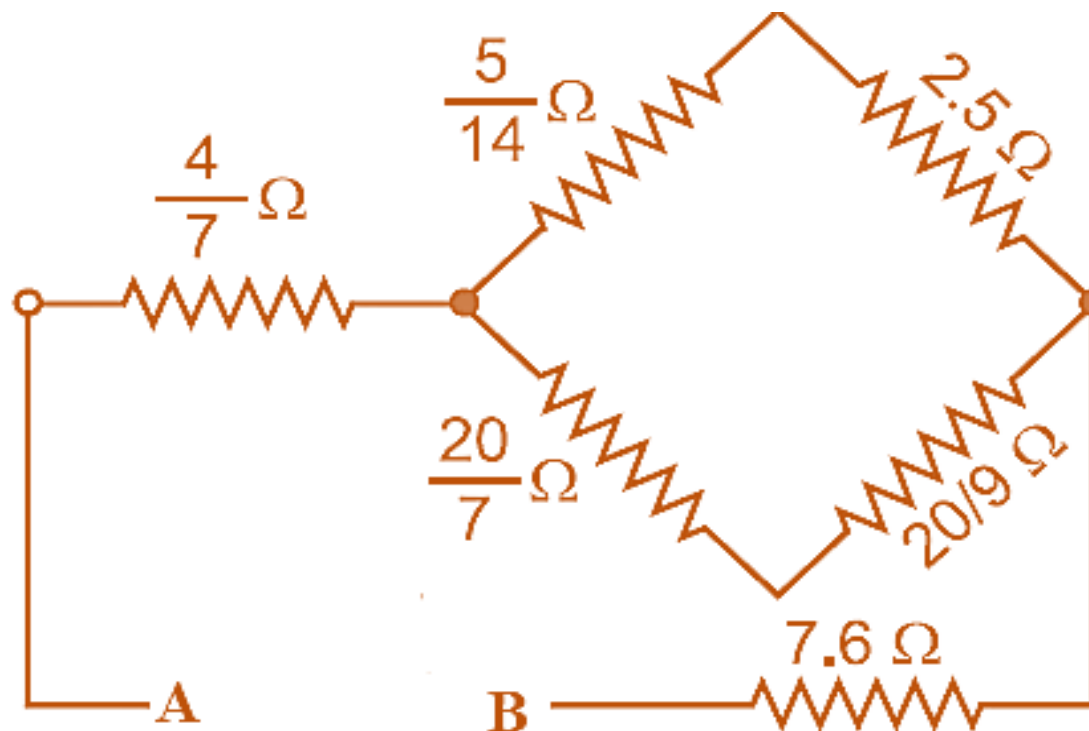
$$R_3' = \frac{8 \times 5}{1 + 5 + 8} = \frac{20}{7} \text{ ohms}$$



# EXERCISE/NUMERICAL ANALYSIS



Circuit now becomes,



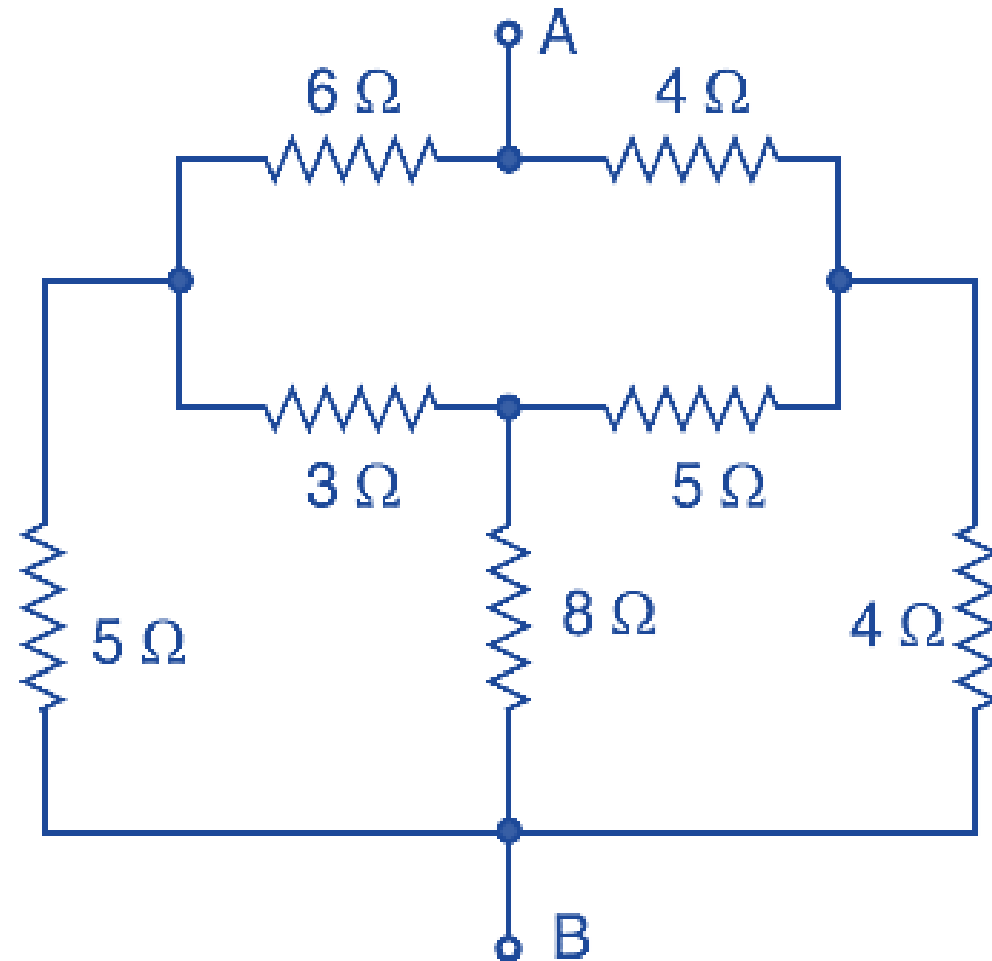
$$R_{AB} = \frac{4}{7} + \left[ \left( \frac{5}{14} + 2.5 \right) \parallel \left( \frac{20}{7} + \frac{20}{9} \right) \right] + 7.6 = 10 \text{ ohms}$$



# EXERCISE/NUMERICAL ANALYSIS



Q. Calculate equivalent resistance across terminals A and B.



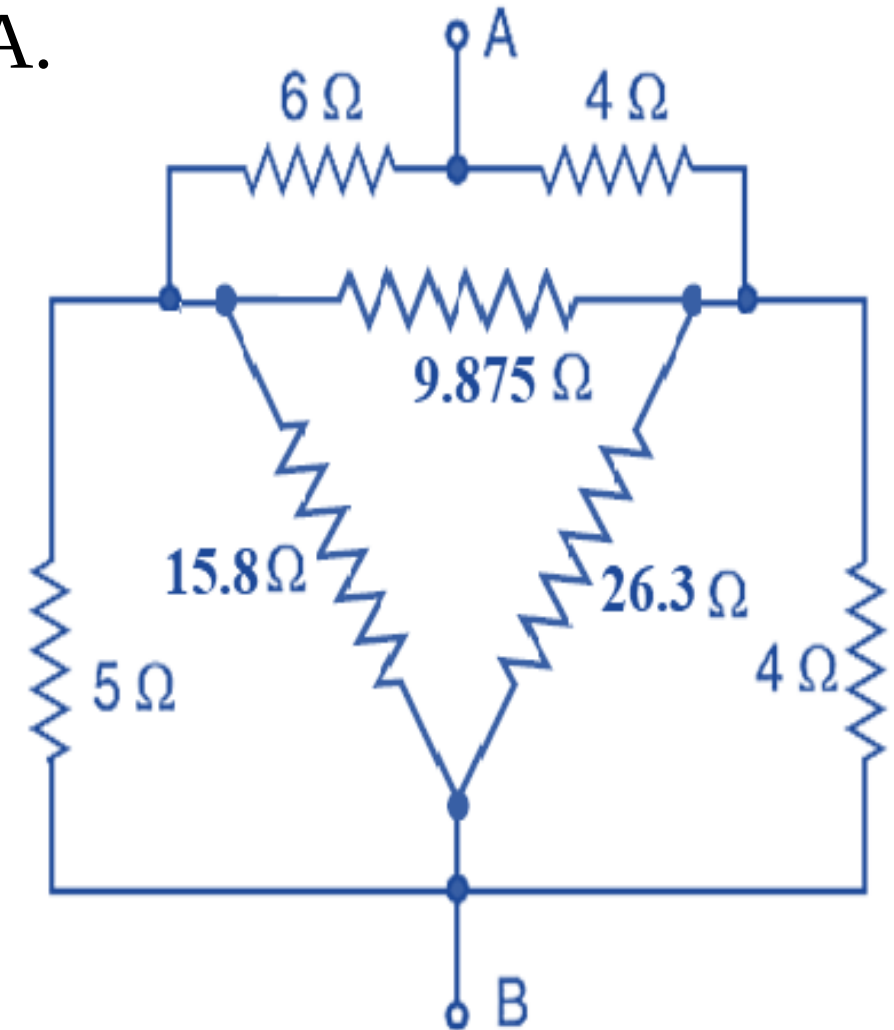
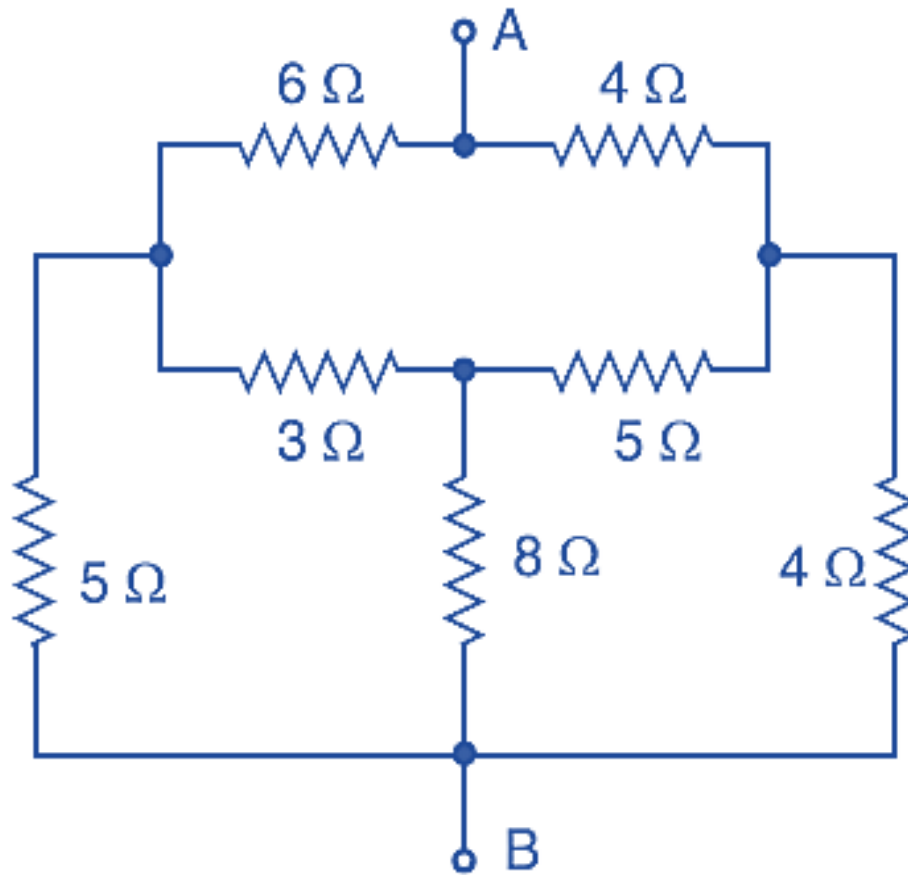




# EXERCISE/NUMERICAL ANALYSIS



Soln: Replacing inner STAR into DELTA.

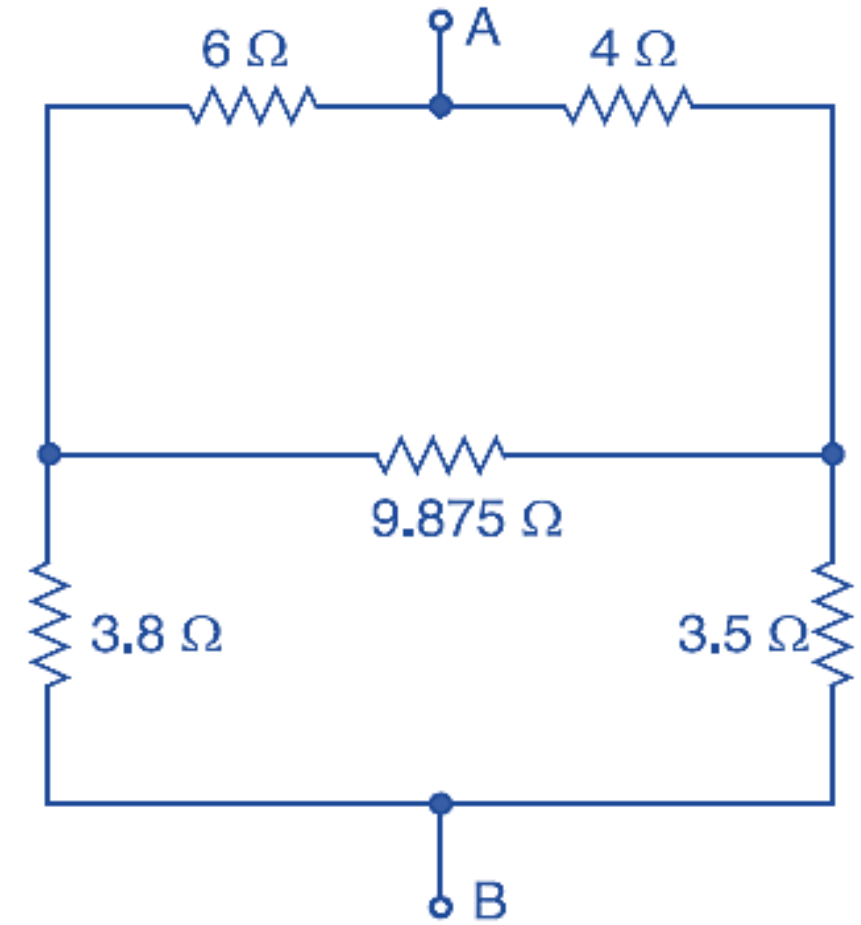
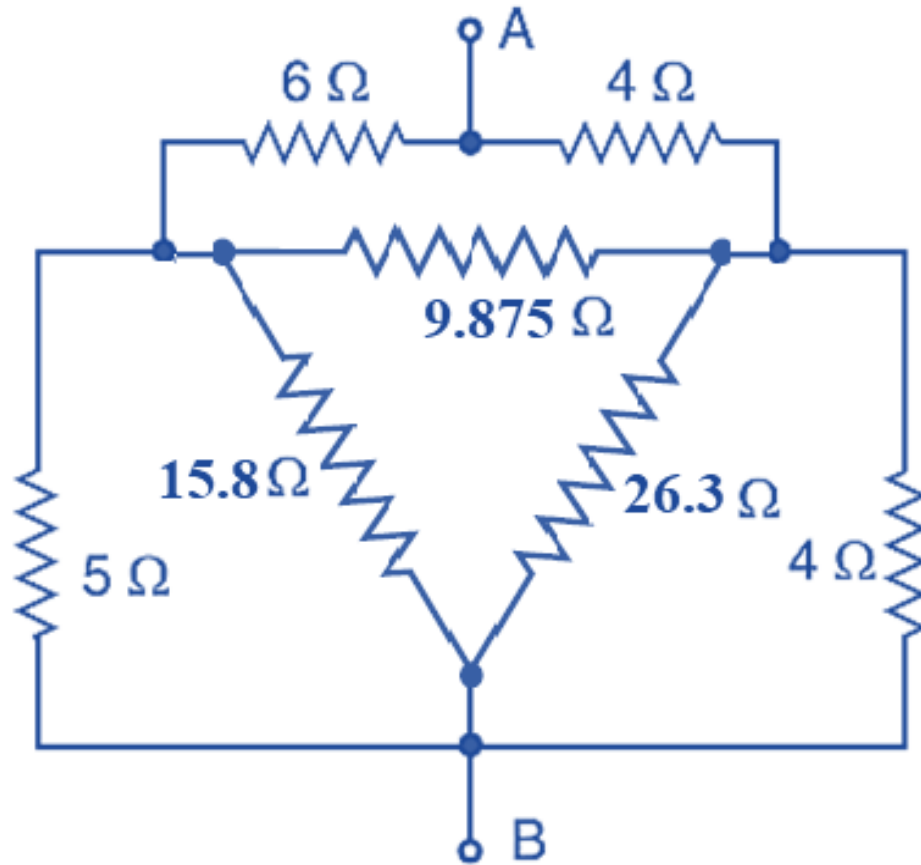




# EXERCISE/NUMERICAL ANALYSIS



15.8 ohm is in parallel with 5 ohm and 26.3 ohm is in parallel with 4 ohm, circuit becomes

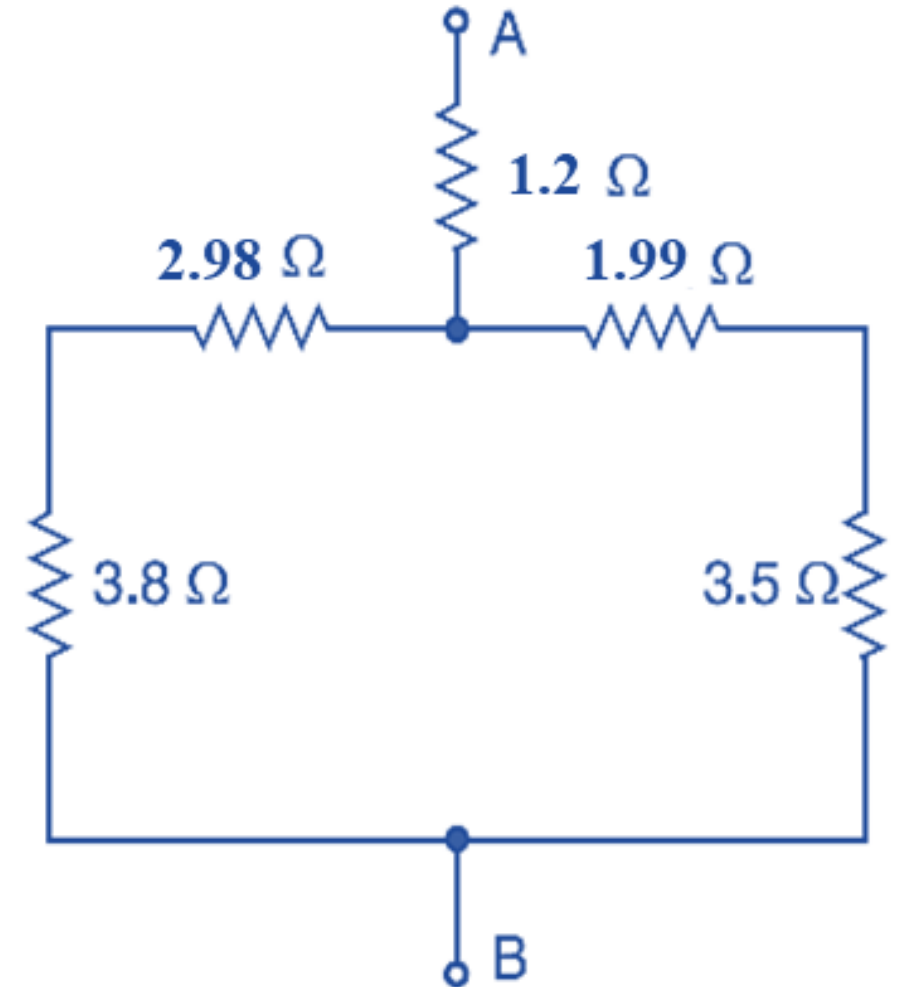
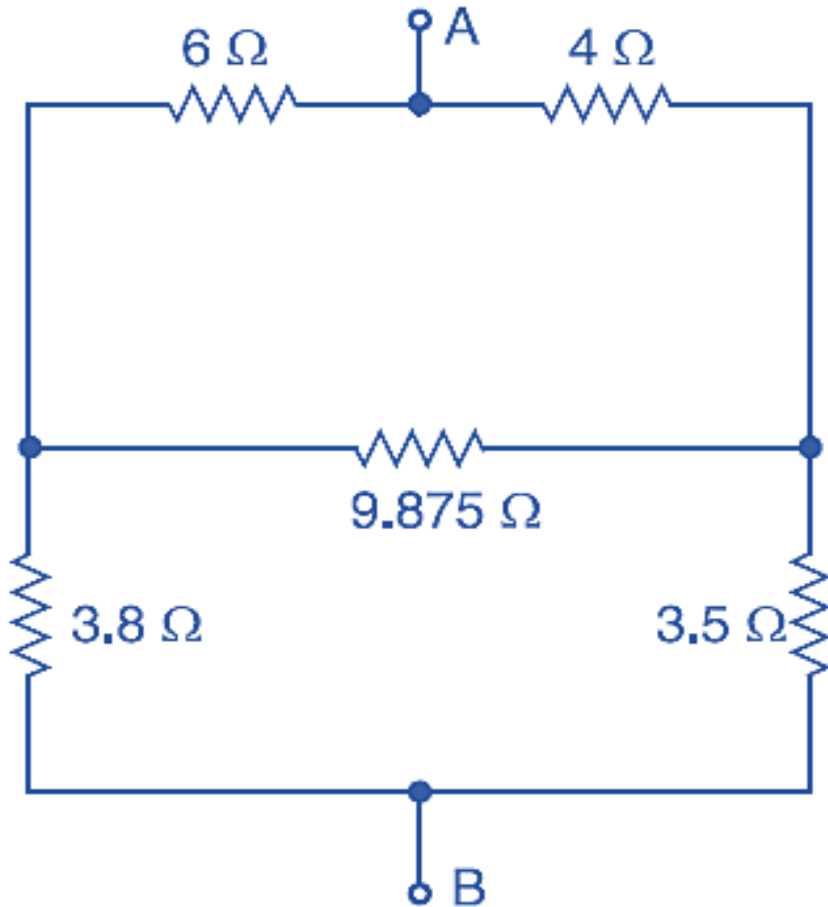




# EXERCISE/NUMERICAL ANALYSIS



Converting upper delta into star,

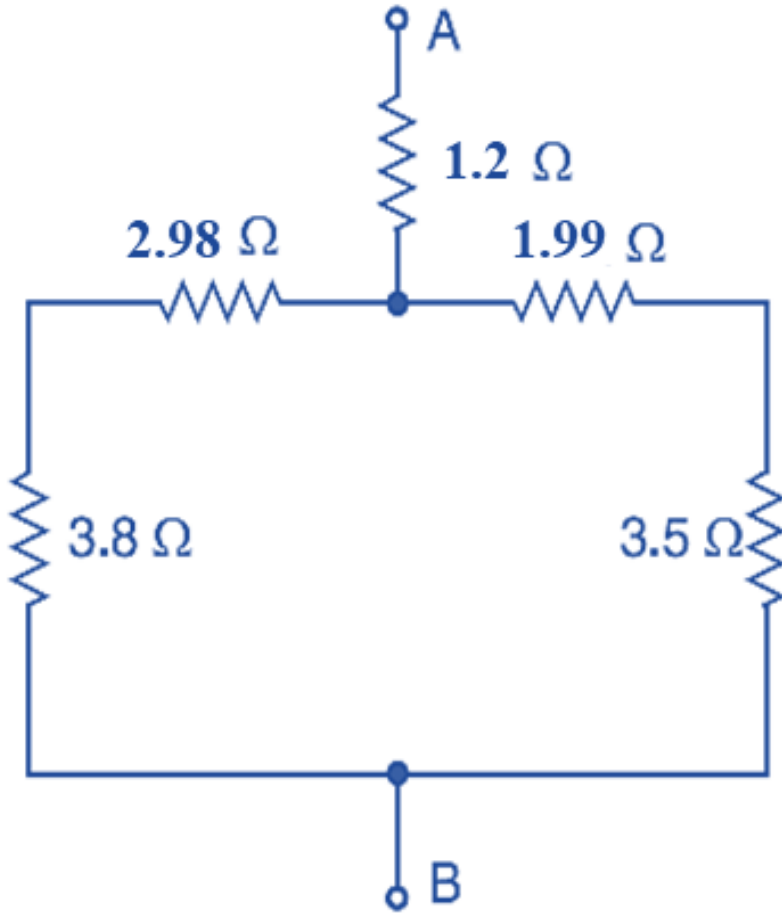




# EXERCISE/NUMERICAL ANALYSIS



Now equivalent resistance can be calculated as,



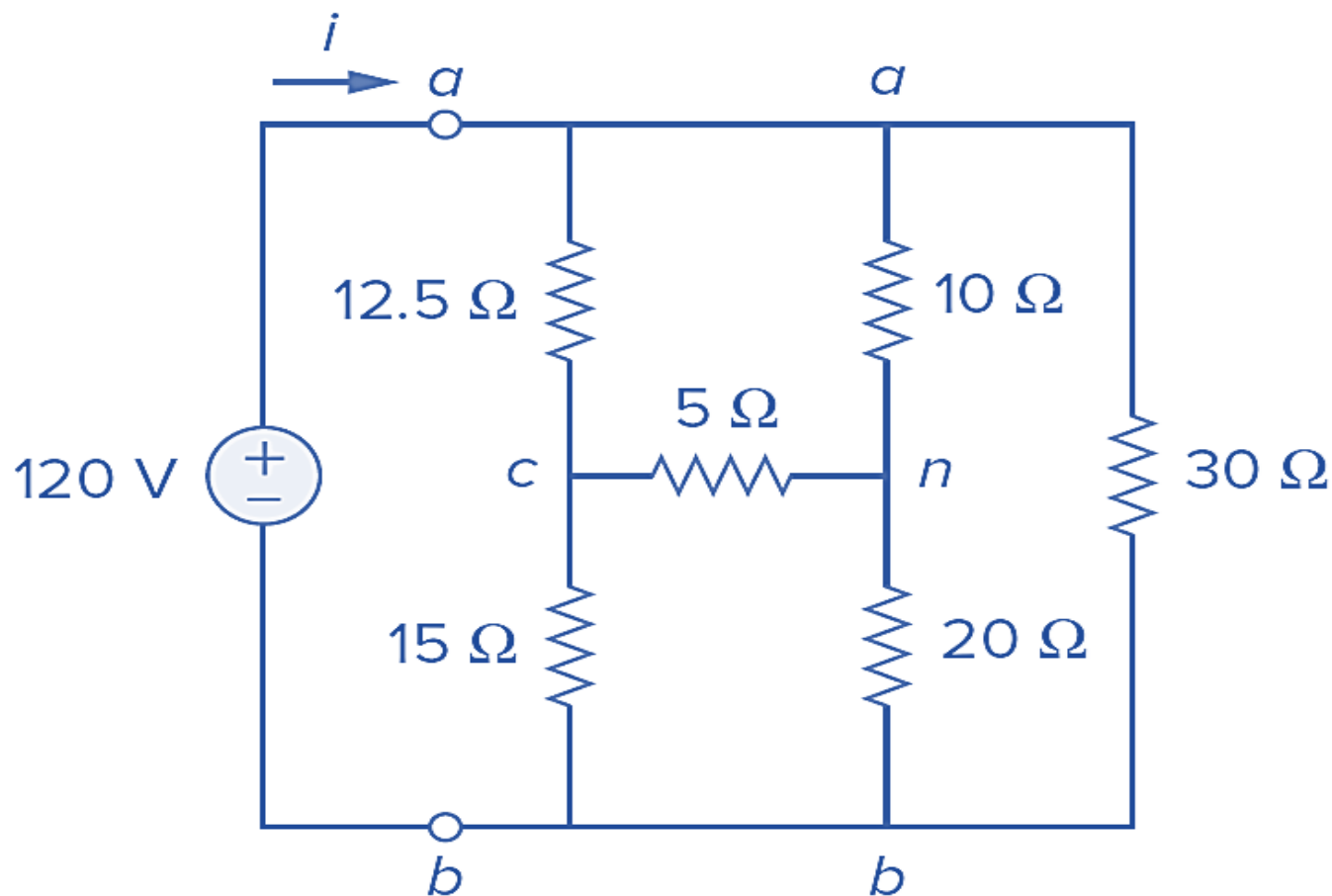
$$R_{eq} = (3.8 + 2.98) \parallel (1.99 + 3.5) + 1.2$$
$$= 4.23 \text{ ohms}$$



# EXERCISE/NUMERICAL ANALYSIS



Q. Obtain the equivalent resistance  $R_{ab}$  for the circuit and use it to find current  $i$ .

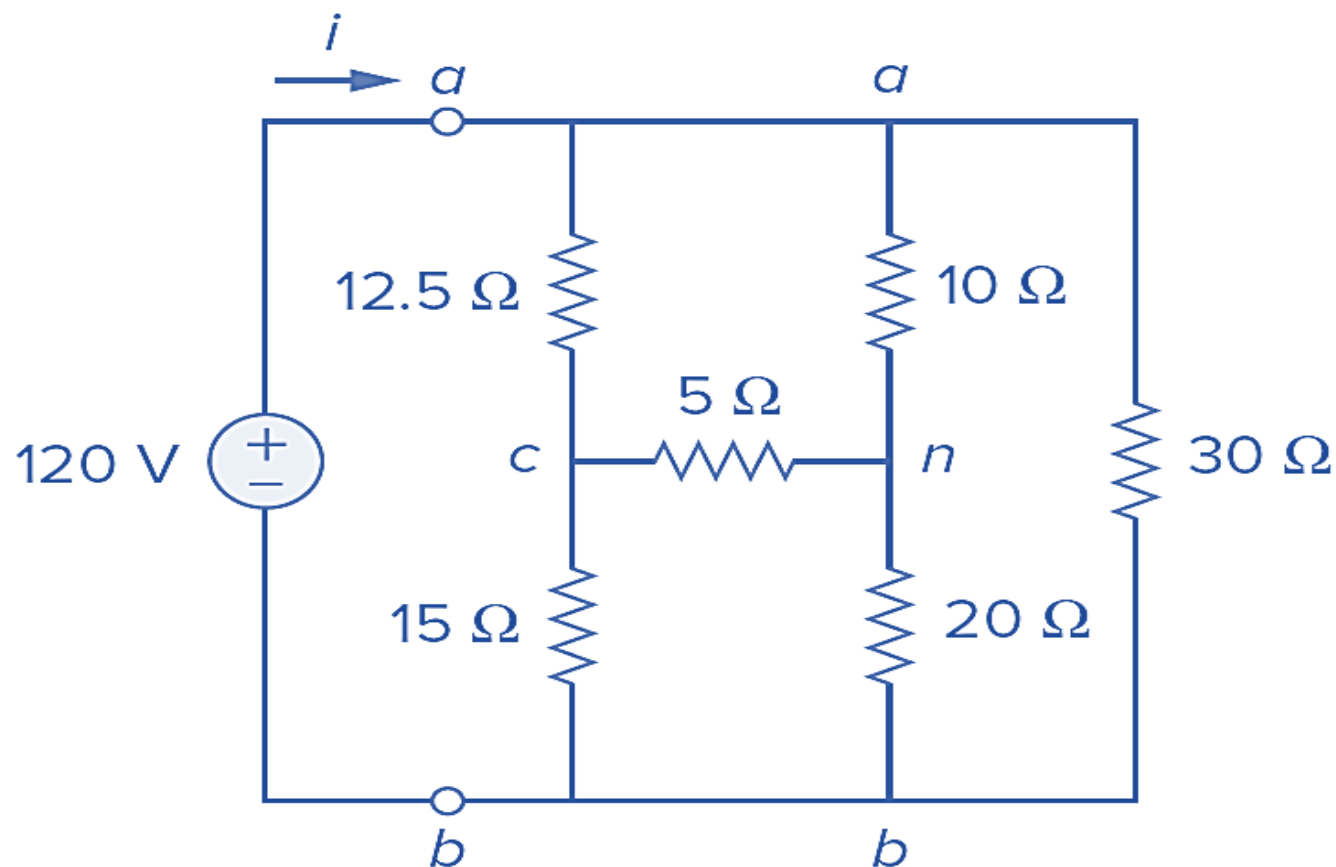




# EXERCISE/NUMERICAL ANALYSIS



Soln: In this circuit, there are two Y networks and three  $\Delta$  networks. Transforming just one of these will simplify the circuit.

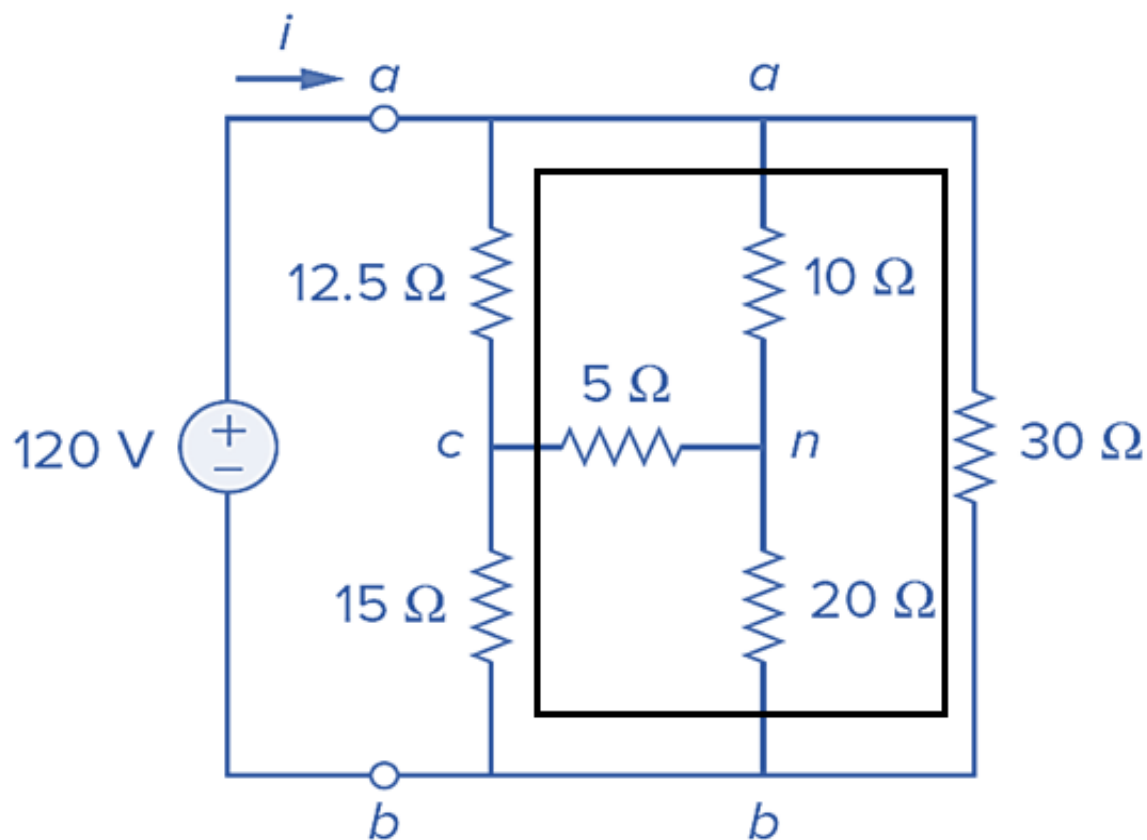




# EXERCISE/NUMERICAL ANALYSIS



We convert the Y-network comprising the 5- $\Omega$ , 10- $\Omega$ , and 20- $\Omega$  resistors into delta.



$$R_1 = 5 + 10 + \frac{5 \times 10}{20} = 17.5 \text{ ohms}$$

(comes in parallel with 12.5  $\Omega$ )

$$R_2 = 5 + 20 + \frac{5 \times 20}{10} = 35 \text{ ohms}$$

(comes in parallel with 15  $\Omega$ )

$$R_3 = 10 + 20 + \frac{10 \times 20}{5} = 70 \text{ ohms}$$

(comes in parallel with 30  $\Omega$ )

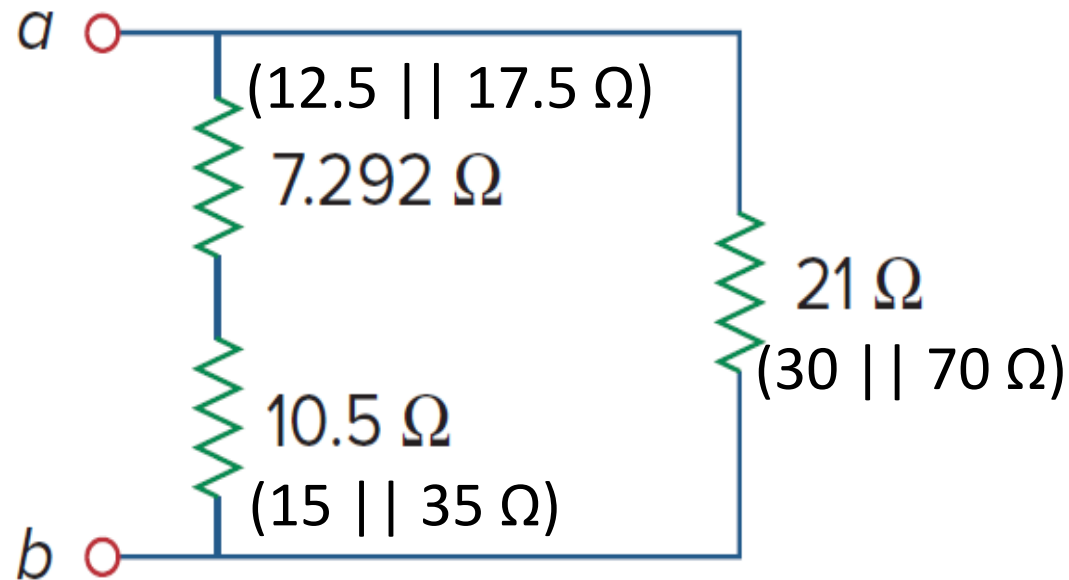




# EXERCISE/NUMERICAL ANALYSIS



Combining the three pairs of resistors in parallel, we obtain.



$$R_{ab} = (7.292 + 10.5) \parallel 21 = \frac{17.792 \times 21}{17.792 + 21} = 9.632 \text{ ohms}$$

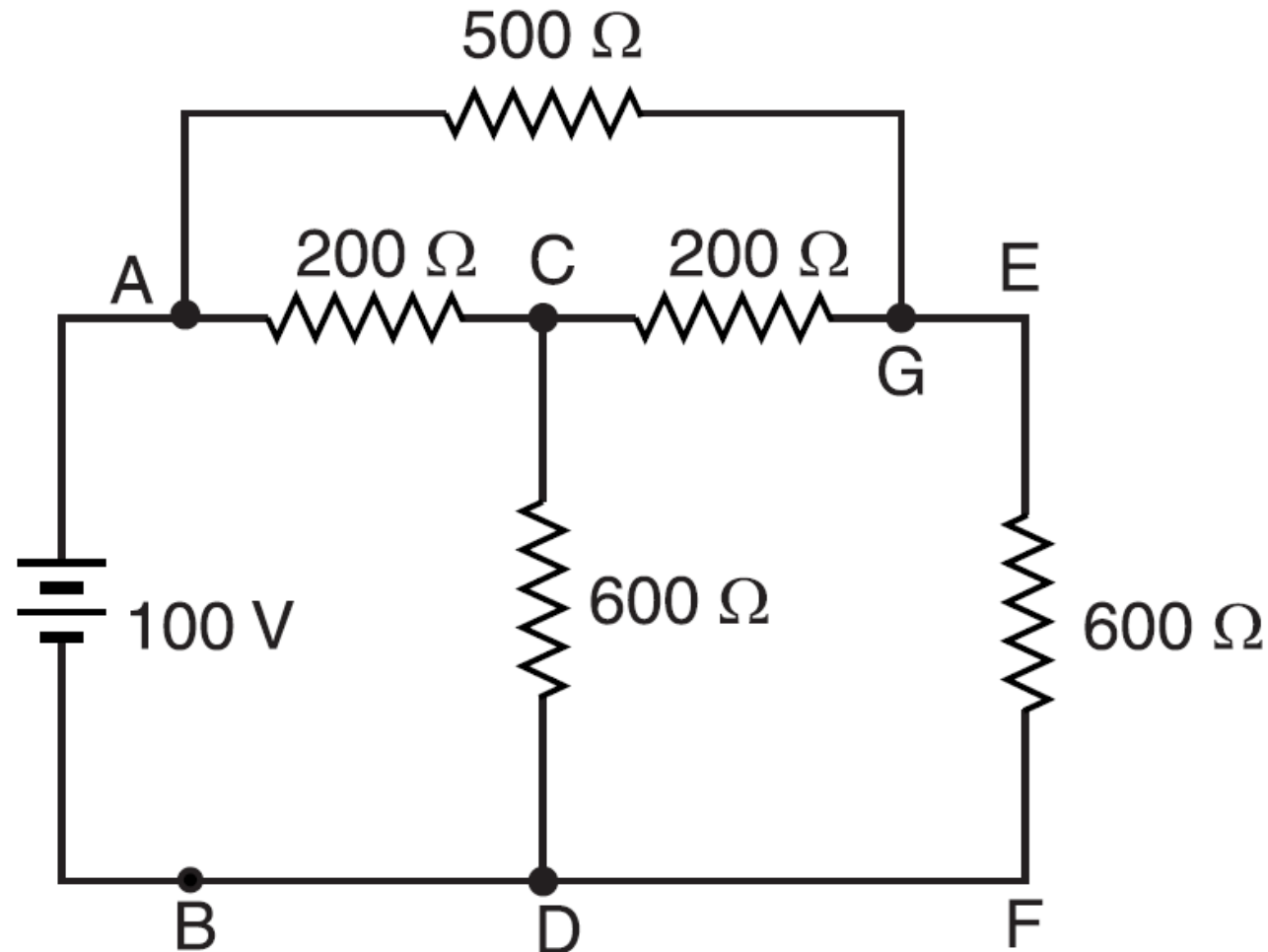
$$i = \frac{v_s}{R_{ab}} = \frac{120}{9.632} = 12.458 \text{ A}$$



# EXERCISE/NUMERICAL ANALYSIS



Q. Determine the load current in branch EF in the circuit shown.

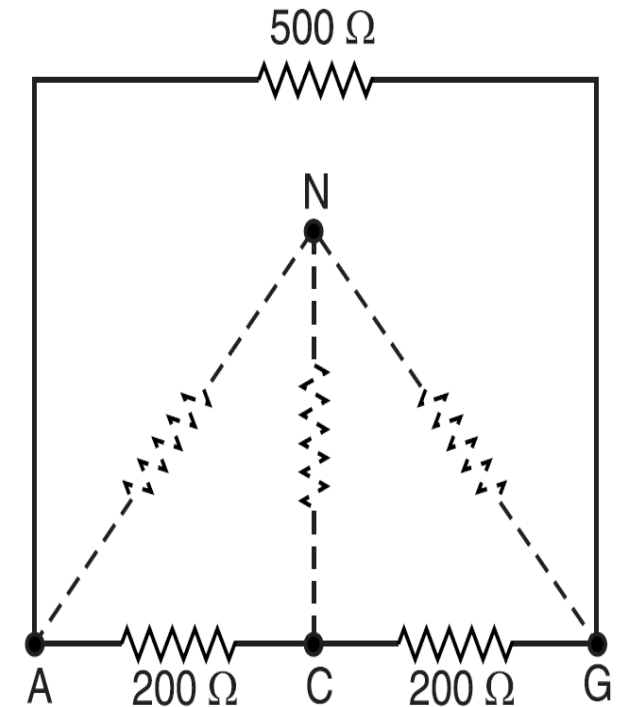
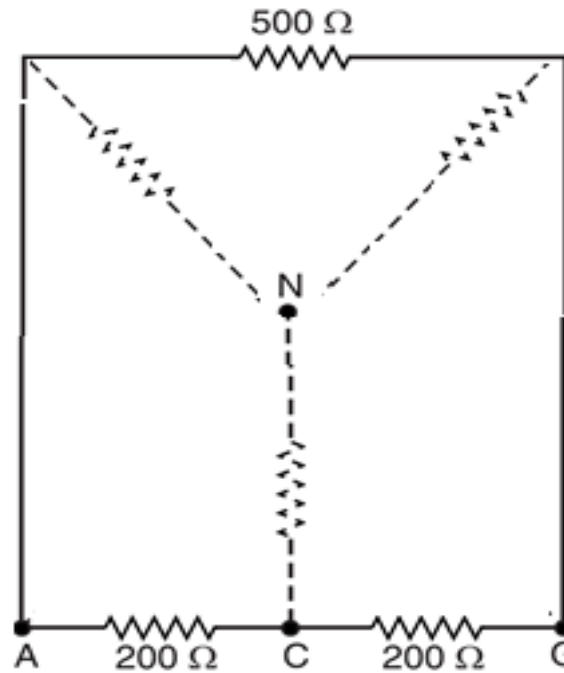
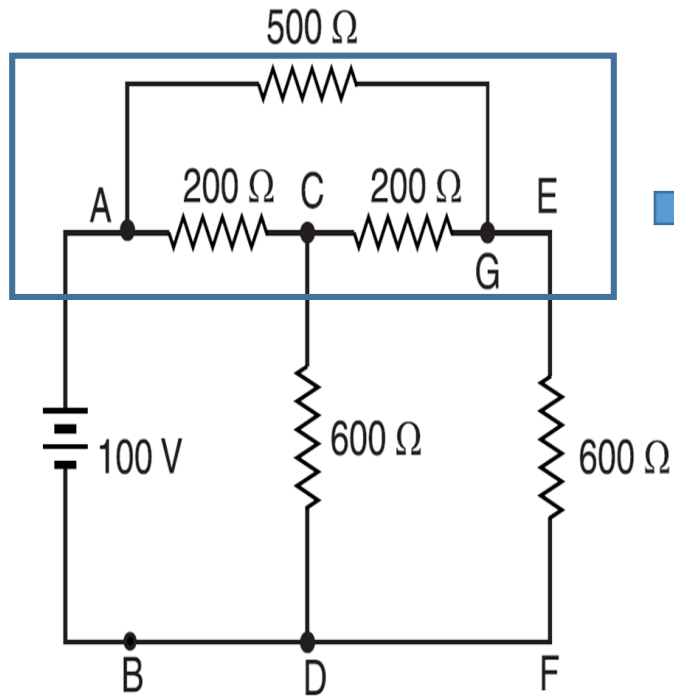




# EXERCISE/NUMERICAL ANALYSIS



Sol. ACGA forms delta, Converting it to equivalent star.



$$R_{AN} = \frac{200 \times 500}{900} = 111.11 \text{ ohms}$$

$$R_{GN} = \frac{500 \times 200}{900} = 111.11 \text{ ohms}$$

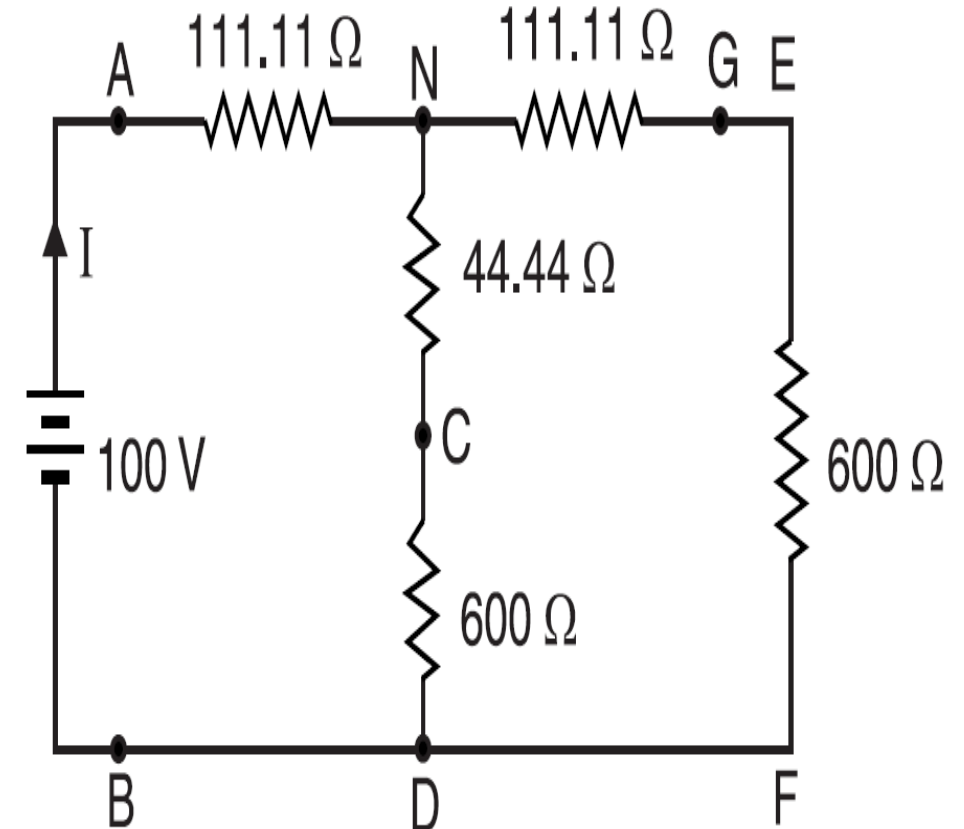
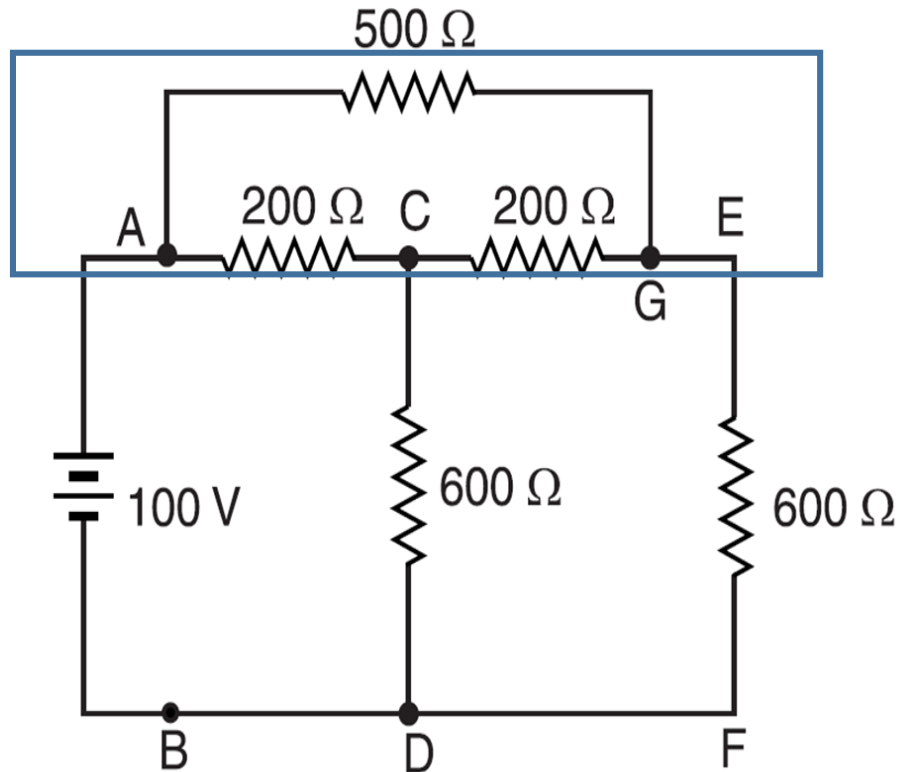
$$R_{CN} = \frac{200 \times 200}{900} = 44.44 \text{ ohms}$$



# EXERCISE/NUMERICAL ANALYSIS



Circuit can be redrawn as



$$R_{NEF} = 111.11 + 600 = 711.11 \text{ ohms}$$

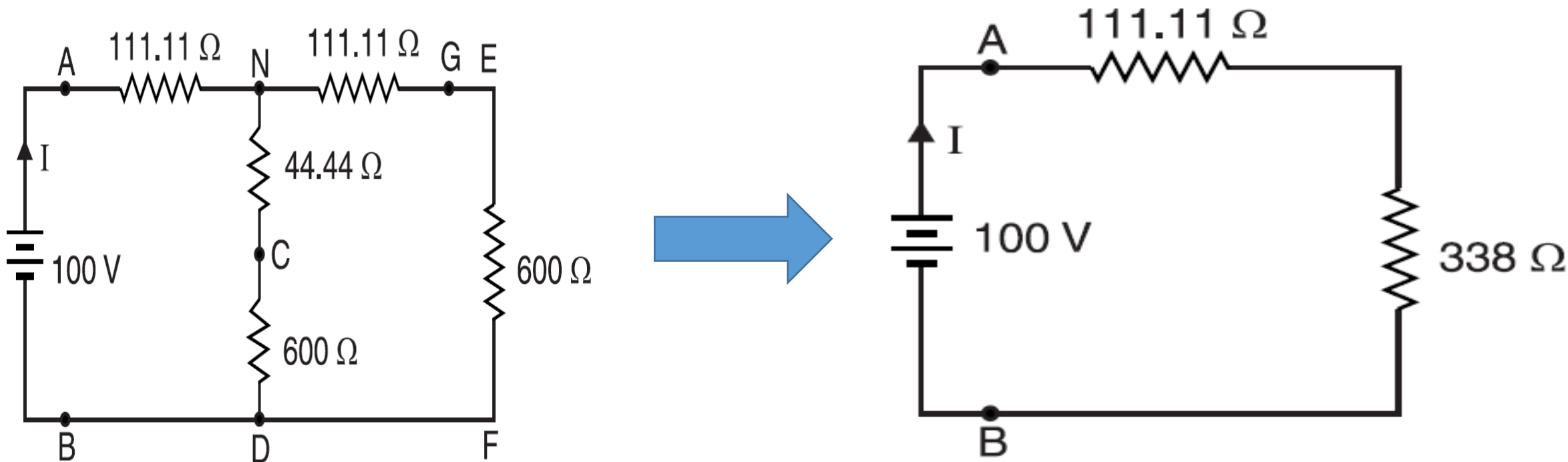
$$R_{ND} = 600 + 44.44 = 644.44 \text{ ohms}$$



# EXERCISE/NUMERICAL ANALYSIS



Branches NCD and NEF are in parallel,  $711.11 \parallel 644.44 = 338$  ohms.



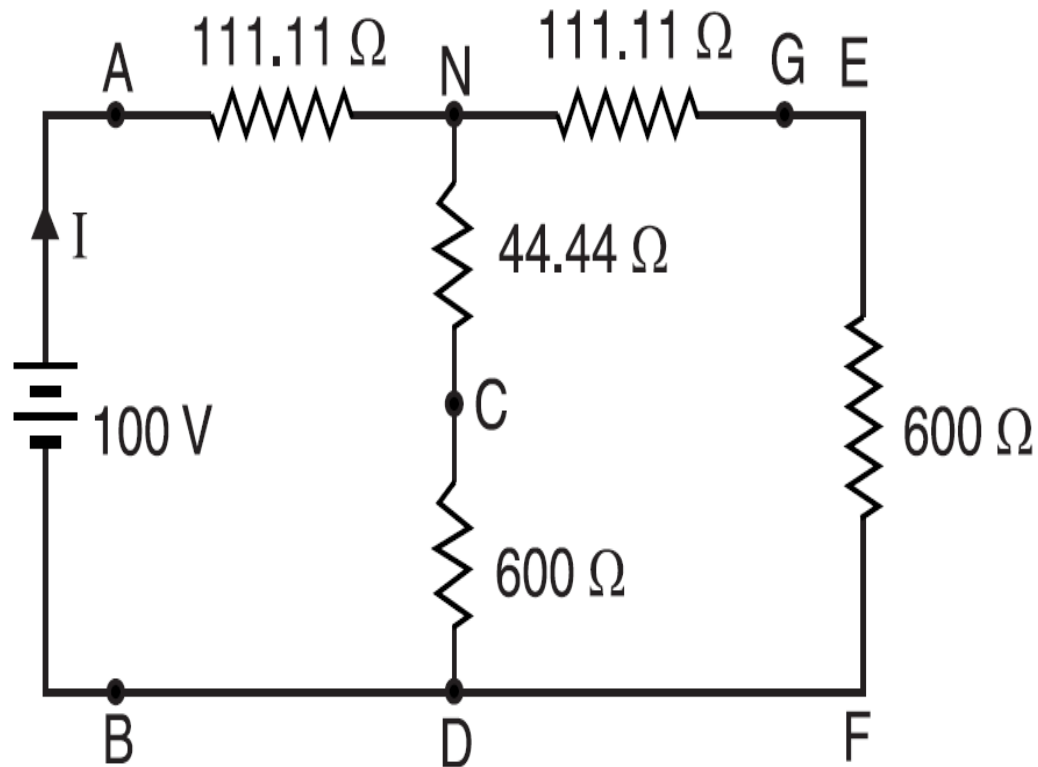
$$\text{Total current } I \text{ in the circuit} = I = \frac{V}{R_{eq}} = \frac{100}{111.11 + 338} = 0.222 A$$



# EXERCISE/NUMERICAL ANALYSIS



To obtain current in branch EF, we apply current division formula.



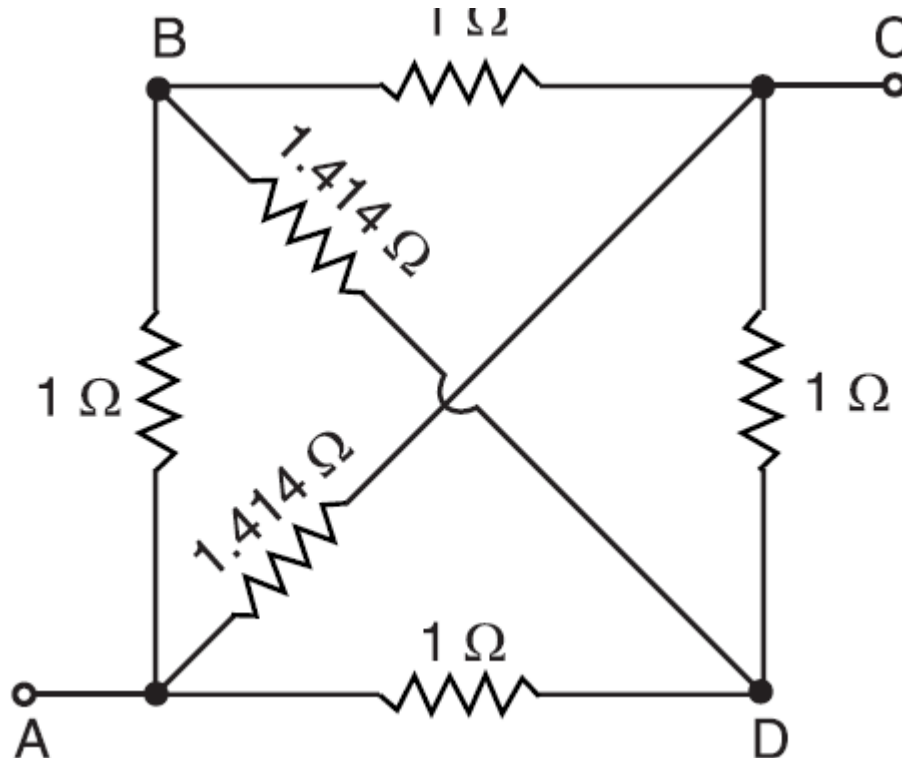
$$\begin{aligned} I_{NEF} &= I \times \frac{R_{NCD}}{R_{NCD} + R_{NEF}} \\ &= 0.222 \times \frac{644.44}{711.11 + 644.44} \\ &= 0.1055A \end{aligned}$$



## EXERCISE



Q. A square and its diagonals are made of a uniform covered wire. The resistance of each side is  $1\ \Omega$  and that of each diagonal is  $1.414\ \Omega$ . Determine the resistance between two opposite corners of the square.



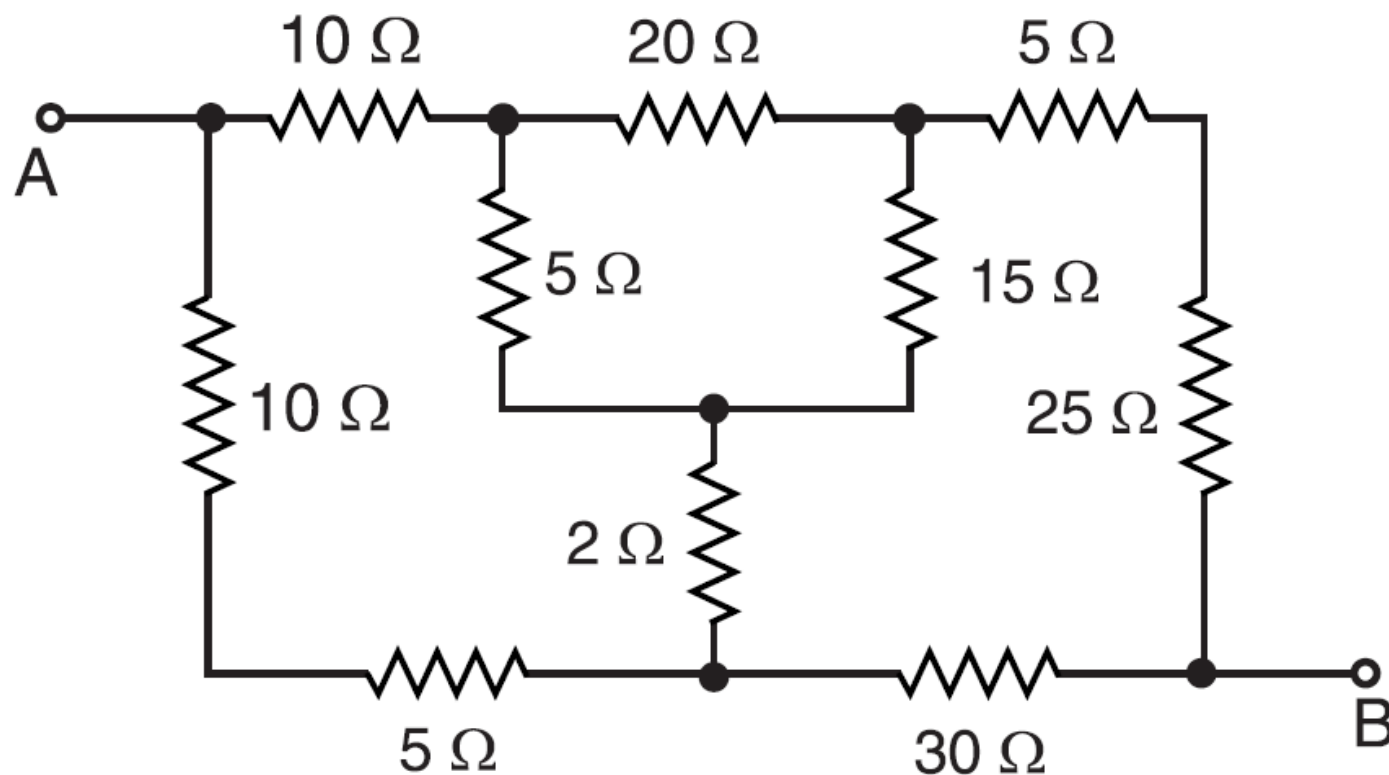




# EXERCISE



Q. Determine the resistance between the terminals A and B of the network.

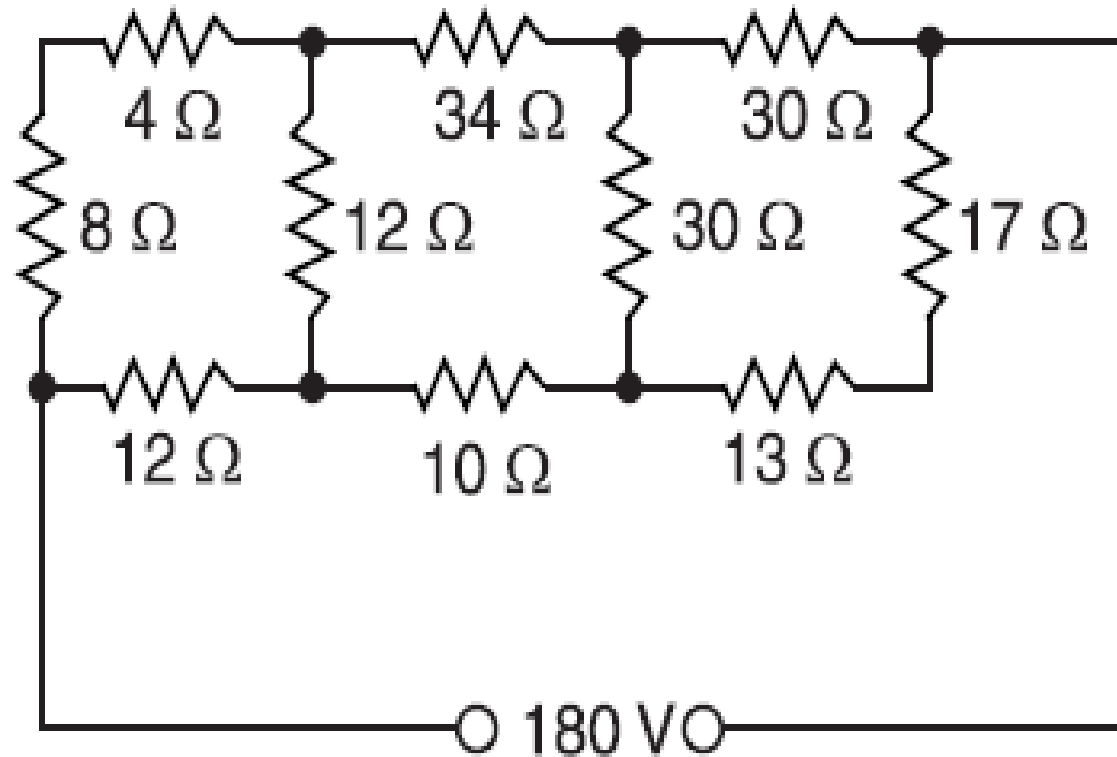




## EXERCISE



Q. Find the current in  $10\ \Omega$  resistor in the network shown by star-delta transformation.

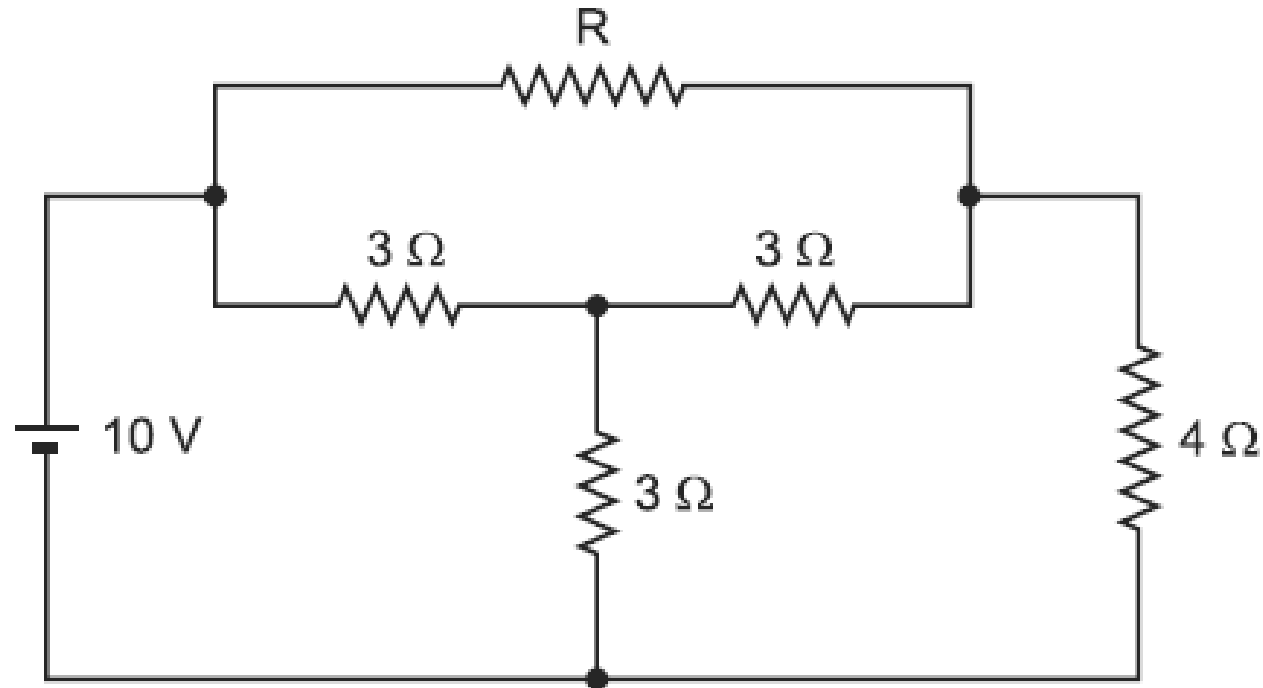




# EXERCISE



Q. Using star/delta transformation, determine the value of  $R$  for the network shown such that  $4\Omega$  resistor consumes the maximum power.



# REFERENCES

- [1] Charles. K. Alexander and Matthew Sadiku “Fundamental of Electric Circuits”, McGraw-Hill Education, 2 Penn Plaza, New York, NY 10121, ch. 2 and 3.
- [2] Edward Hughes, John Hiley, Keith Brown and Ian McKenzie Smith Hughes “Electrical & Electronic Technology”, Pearson Education Limited, Edinburgh Gate, England, ch. 2 and 3.
- [3] V. K. Mehta and Rohit Mehta “Basic Electrical Engineering”, S. Chand & Company Pvt. Ltd., Ram Nagar, New Delhi, ch. 2.